The Effective-Support Floor of the Adversarial Shadow Price: An $\Omega(\sqrt{k_\varepsilon})$ Lower Bound

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Abstract

Adversarial robustness is a one-dimensional resource with a price. We treat the adversarial radius ε as a one-dimensional budget of a constrained value function $L^*(\mathcal{E})$ and define the *marginal price of robustness* as its subgradient $\lambda_\varepsilon = \partial L^*/\partial\varepsilon$. Our main result is a *floor on this price*, whose correct pricing unit is the *effective support* of the robust-optimal discriminant, $k_\varepsilon := (\|w^\star\|_1/\|w^\star\|_2)^2$ (the participation ratio of the soft-thresholded active support): at a near-threshold operating point,

$$\lambda_\varepsilon \;\geq\; c_M \sqrt{k_\varepsilon}/\sigma,$$

with the ℓ_∞ adversary and ℓ_1 duality as its geometric origin. This k_ε-floor is *unconditional* in a Gaussian–linear regime (Theorem 4.1) and, distribution-free, for all symmetric log-concave noise (Theorem 5.5); it prices $w^\star$ by its own effective support and is not, by itself, a dimensional bound. Its *dimensional* specializations are conditional: the dense case $k_\varepsilon = \Theta(d)$ giving $\Omega(\sqrt{d})$ holds under a checkable delocalization condition (Proposition 5.6, unconditional for Gaussian), the sparse case gives $\Omega(\sqrt{k})$ under support delocalization, and the stable-rank case $k_\varepsilon = \Theta(r_{\mathrm{eff}})$ ($r_{\mathrm{eff}} = \mathrm{tr}(\Sigma)/\|\Sigma\|$) giving $\Omega(\sqrt{r_{\mathrm{eff}}})$ holds under aligned effective occupancy (Proposition 6.3, conditional). The proportional-asymptotic sharp constant is supplied by the convex Gaussian min–max theorem (CGMT; Proposition 5.1, literature-derived), and the floor transfers to the local operating point of any C^2 classifier through an ℓ_∞ quadratic-form curvature κ_∞ (Proposition 7.1). Synthetic experiments confirm that $\sqrt{k_\varepsilon}$ is the universal pricing quantity across the dense, sparse, and low-effective-rank regimes, and that the conditions of the stable-rank specialization are necessary.

Keywords: adversarial robustness; shadow price; effective support; ℓ_∞/ℓ_1 duality; stable rank; convex Gaussian min–max theorem.

1 Introduction

The cost center of large models is shifting. What used to be a one-time pre-training expense is increasingly incurred at inference time—longer context windows, longer reasoning chains, multi-step agents, multimodal alignment—so that "compute" is now a recurring cost accrued at deployment as much as at training. A parallel one-dimensional cost accrues with *deployment-time robustness*: as models are asked to stay consistent under adversarial input perturbations, keeping their behavior invariant demands a nontrivial training and certification price. Deciding how much of this budget to spend is therefore no longer a one-shot allocation made before training; it is a recurring, cross-cutting pricing problem, and the dimension of that problem that is hardest to quantify—and easiest for practitioners to underestimate—is the adversarial budget.

This paper focuses on that dimension. We ask a concrete question: viewed as the subgradient of a value function, where does the size of the marginal price of robustness $\lambda_\varepsilon = \partial L^*/\partial\varepsilon$ come from? Does it grow with the ambient dimension d? When can it be replaced by an intrinsic dimension of the representation? The answer is nontrivial. In high-dimensional deep models, even under the optimal discriminant, the marginal cost of one unit of adversarial radius cannot be made arbitrarily small: there is an *effective-support floor* forced by the ℓ_∞ adversary and the ℓ_1 dual geometry. When the robust-optimal direction spreads densely across d coordinates, this floor degenerates to $\Omega(\sqrt{d})$. This is, we argue, the most fundamental source of the observation that high-dimensional models are "economically hard to robustify."

Contributions. We make the floor rigorous and identify its correct pricing unit.

(i) We write resource-constrained learning as a value function $L^*(\mathcal{E})$ whose subgradient with respect to the budget is the shadow price $\lambda = -\partial L^*/\partial\mathcal{E}$; the adversarial radius ε is one such budget, with price $\lambda_\varepsilon \in \partial_\varepsilon L^*$ (Section 2).

(ii) We establish the structural role of a *readout operator* Π_Y: the robust marginal price, the training-side costate, and static-budget gradients are three readouts of the same object ∂L^* along different budget coordinates; and we give a local bridge (Proposition 8.1, $\varepsilon \to 0$) that connects the robust price to a training-side input gradient.

(iii) Section 2 gives the nonsmooth foundation on which the price lives: Rademacher's theorem guarantees that the price is almost-everywhere single-valued, degenerating to a Danskin interval on a measure-zero critical set.

(iv) Section 4 is the core. Theorem 4.1 shows that the correct pricing unit of the robust marginal price is the effective support k_ε (the participation ratio of the soft-thresholded active support $(|\mu| - \varepsilon)_+$): at a near-threshold (margin-density) operating point, $\lambda_\varepsilon \geq c\sqrt{k_\varepsilon}/\sigma$. Dense signals give $k_\varepsilon = \Theta(d)$ and hence the $\Omega(\sqrt{d})$ special case (with a population absolute constant; the proportional-asymptotic sharp constant is supplied by the CGMT, Proposition 5.1). Proposition 6.3 replaces k_ε by the stable rank $r_{\text{eff}} = \operatorname{tr}(\Sigma)/\|\Sigma\|$ (a distinction we return to below), yielding $\lambda_\varepsilon \geq c\sqrt{r_{\text{eff}}}/\sqrt{\|\Sigma\|}$ under an effective-occupancy condition (conditional); the adaptive-feature version is an open problem we state explicitly.

We stress a distinction that runs through the paper: k_ε is a geometric quantity in the *discriminant weight space*, whereas $r_{\text{eff}} = \operatorname{tr}(\Sigma)/\|\Sigma\|$ is a geometric quantity in the *data covariance space*. Under signal–noise co-support the two coincide—the effective support in weight space collapses onto the stable rank in data space—which is exactly what Proposition 6.3 makes precise.

Main Mechanism. ℓ_∞ attack $\Rightarrow$ per-coordinate margin shrinkage $-\varepsilon\|w\|_1$ $\Rightarrow$

$$\lambda_\varepsilon = \frac{\|w^\star\|_1}{\sigma\|w^\star\|_2}\,\phi\big(t(\varepsilon)\big) = \frac{\sqrt{k_\varepsilon}}{\sigma}\,\phi\big(t(\varepsilon)\big), \qquad \text{and} \qquad |t| \leq M \;\Rightarrow\; \lambda_\varepsilon \geq c_M\frac{\sqrt{k_\varepsilon}}{\sigma}.$$

The operational dimension of adversarial pricing is the effective support k_ε; the ambient dimension d is only its dense special case, and $\sqrt{k_\varepsilon}$ is the unified pricing quantity throughout (Theorems 4.1, Corollary 4.4, Proposition 6.3).

Claim status. Table 1 summarizes the status and key assumptions of each result. We use *Established* for self-contained proofs, *Literature-derived conditional* for results obtained by invoking a published CGMT characterization and specializing it, *Conditional* for results resting on explicit hypotheses, and *Conjectural* for open problems.

Roadmap. Section 2 builds the formal foundation of the adversarial shadow price: the value function, ε as a (tightening) adversarial budget, the KKT/envelope structure, the readout operator, and the nonsmooth foundation. Section 3 gives the ℓ_∞/ℓ_1 geometry and the closed-form robust risk (Lemma 3.1). Section 4 establishes the population effective-support floor (Theorem 4.1), with dense and sparse special cases. The proportional-asymptotic sharp constant (CGMT) and the distribution-free extension to symmetric log-concave noise (Theorem 5.5), the effective-rank refinement, the local deep-network extension, the implications and synthetic validation, related work, and conclusions occupy the remaining sections.

Result	Statement	Key assumptions	Status
Lemma 3.1	Closed-form ℓ_∞/ℓ_1 robust risk and price	Gaussian noise $z \sim N(0, \sigma^2 I)$, linear discriminant, binary	Established
Theorem 4.1	Effective-support floor $\lambda_\varepsilon \geq c_M\sqrt{k_\varepsilon}/\sigma$	Gaussian–linear; near-threshold margin-density $\lvert t\rvert \leq M$	Established
Theorem 5.5	Same k_ε-floor, distribution-free (not a $\sqrt{d}$ bound)	Symmetric log-concave noise + margin-density/operating-CDF level $\geq \alpha_0$	Established
Corollary 4.4	Sparse $\sqrt{k}$ lower bound	Theorem 4.1 + k-sparse μ with active participation $\Theta(k)$ (population)	Established
Prop. 5.1	Proportional-asymptotic sharp constant (CGMT)	Gaussian mixture, $n/d \to \kappa$, convex surrogate + ridge	Literature-derived cond.
Prop. 6.3	$k_\varepsilon = \Theta(r_{\text{eff}})$ under aligned effective occupancy	(EO0) basis alignment; (EO1) flat subspectrum; (EO2) signal-tail control; (EO3) effective-signal delocalization; (EO4) subgradient-tail control; non-adaptive	Conditional
Prop. 7.1	Local first-order deep-net sensitivity	C^2 classifier, working point, first-order neighborhood, κ_∞ remainder	Established

Table 1: Claim status and key assumptions (core results; the full table appears with the proportional-asymptotic and effective-rank sections).

2 The Adversarial Shadow Price

2.1 Learning under a resource constraint

We write a resource-constrained learning problem as a constrained extremal value function. Let X be a resource–control space (containing a training-control subspace and various static-budget axes, together with the adversarial-radius axis), and let $L^*\colon X \to \mathbb{R}$ be the value function

$$L^*(\mathcal{E}) = \min_{x \in X} \left\{ L(x) \ : \ g_i(x) \leq \mathcal{E}_i, \ i = 1, \ldots, d \right\},$$

where L is the loss, g_i the resource constraints, and $\mathcal{E} = (\mathcal{E}_1, \ldots, \mathcal{E}_d)$ the budget. The adversarial radius ε is one of these budgets, entering as a *tightening* constraint: a larger ε shrinks the feasible set and raises L^*.

2.2 The variational principle

Under standard regularity we obtain the multipliers whose one of whose coordinates is the shadow price of the adversarial budget.

Theorem 2.1 (d-dimensional resource variational principle; existence and uniqueness of multipliers)**.** *Assume* (A1) *convexity (X convex, L and each g_i convex and C^1, at least locally near the optimum) and* (A2) *a constraint qualification (Slater or MFCQ; see Bertsekas, 1999). Then the optimal allocation $x^*(\mathcal{E})$ satisfies the KKT stationarity conditions: there exist multipliers $\lambda = (\lambda_1, \ldots, \lambda_d) \geq 0$ (in general not unique) such that*

$$\nabla L(x^*) + \textstyle\sum_i \lambda_i \nabla g_i(x^*) = 0, \qquad \lambda_i\big(g_i(x^*) - \mathcal{E}_i\big) = 0, \qquad i = 1, \ldots, d.$$

If in addition (A3) *LICQ (the active-constraint gradients $\{\nabla g_i(x^*) : g_i(x^*) = \mathcal{E}_i\}$ are linearly independent) or* (A3$'$) *strict complementarity together with a second-order sufficient condition hold, then λ is unique. Wherever the paper needs the price to be single-valued (the λ_ε expression of Section 4 and the ε^{**} first-order condition), we restrict to a neighborhood on which L^* is differentiable; by Proposition 2.2 this is equivalent to requiring λ to be unique there, and by Rademacher's theorem it holds for Lebesgue-a.e. ε.*

Complementary slackness reads economically: only a fully used resource carries a positive price.

2.3 The shadow price as the gradient of the value function

Proposition 2.2 (shadow price as the (sub)gradient of the value function)**.** *Under the assumptions of Theorem 2.1: (a) at a differentiable point (equivalently, where the multiplier is unique, by (A3) or (A3$'$))*

one has the strict envelope identity $\lambda_i = -\partial L^*/\partial \mathcal{E}_i$; (b) in general L^* is locally Lipschitz but need not be differentiable everywhere, and $-\lambda$ is an element of the Clarke subdifferential $\partial_{\mathcal{E}} L^*$. In particular, the adversarial price λ_ε is single-valued exactly where L^* is differentiable in the ε direction; on the critical set it degenerates to a Danskin interval (Section 2, nonsmooth foundation).

Thus ∇L^* splits into a *direction* (which budget to relax) and a *magnitude* (the price); the readout operator below reads it along different coordinates.

Corollary 2.3 (marginal equality). *At the optimum, and under a total budget $\sum_i \mathcal{E}_i \le B$, reallocating the budget equalizes the marginal loss reduction purchased per unit of budget across active dimensions. This is the classical envelope condition at the optimum of the value function.*

2.4 The readout operator (deferred)

The shadow price is a single object ∂L^* read along different budget coordinates: the training coordinate gives the Pontryagin costate, a static coordinate an ordinary gradient, the adversarial-radius coordinate the Clarke subdifferential—three readouts of one object, not three theories. We use two consequences: the floor of Section 4 is the readout along ε (independent of which readout is taken), and as $\varepsilon \to 0$ the robust price is estimated by a training-side gradient norm $\mathbb{E}\|\nabla_x \ell\|_*$ (Proposition 8.1). The formal readout operator Π_Y and the "two readouts" identity are deferred to Appendix A.4.

2.5 Nonsmooth foundation

The measure-theoretic phrases above (a.e. single-valued, Danskin interval on the critical set) require the nonsmooth apparatus. For a locally Lipschitz $f\colon \mathbb{R}^n \to \mathbb{R}$, the Clarke subdifferential $\partial f(x) = \operatorname{conv}\{\lim \nabla f(x_k) : x_k \to x\}$ (the convex hull of nearby gradient limits) coincides with the convex subdifferential when f is convex; the robust marginal price proper is $\lambda_\varepsilon \in \partial_\varepsilon L^*(\mathcal{E})$, and the equality $\lambda_\varepsilon = \partial L^*/\partial\varepsilon$ holds only at differentiable points. By Rademacher's theorem (a locally Lipschitz function is Lebesgue-a.e. differentiable), for a.e. ε the set $\partial_\varepsilon L^* = \{\lambda_\varepsilon\}$ is a singleton and the price is a well-defined classical derivative; nonsmoothness is confined to a Lebesgue-null set.

3 The ℓ_∞/ℓ_1 Geometry

We now specialize to the setting in which the floor is proved. Take the standard binary model $x = y\mu + z$ with label $y \in \{\pm 1\}$ equiprobable and noise $z \sim N(0, \sigma^2 I_d)$, and a linear discriminant $x \mapsto \operatorname{sign}\langle w, x\rangle$. Against an ℓ_∞-bounded adversary $\|\delta\|_\infty \le \varepsilon$, the worst-case perturbation reduces the margin by exactly $\varepsilon\|w\|_1$, which is the signature of the ℓ_∞ attack and the origin of the ℓ_1 dual geometry.

Lemma 3.1 (closed-form robust risk and price). *For the model above and a linear discriminant w, the robust (ℓ_∞, radius ε) misclassification risk is*

$$R(\varepsilon; w) = \Phi\big(t(w)\big), \qquad t(w) := \frac{\varepsilon\|w\|_1 - \langle w, \mu\rangle}{\sigma\|w\|_2},$$

which is monotonically increasing in ε. Its marginal price is

$$\lambda_\varepsilon(w) = \frac{\partial R}{\partial \varepsilon} = \frac{\|w\|_1}{\sigma\|w\|_2}\,\phi\big(t(\varepsilon)\big) = \frac{\sqrt{k_\varepsilon(w)}}{\sigma}\,\phi\big(t(\varepsilon)\big), \qquad k_\varepsilon(w) := \Big(\tfrac{\|w\|_1}{\|w\|_2}\Big)^2,$$

where $\phi = \Phi'$ is the standard normal density evaluated at the standardized margin $t(\varepsilon)$. (Proof in Appendix B.1.)

Thus, already at the level of a fixed discriminant, the price factors into a purely geometric quantity $\sqrt{k_\varepsilon(w)} = \|w\|_1/\|w\|_2$ (the square root of the participation ratio of w) and a density factor $\phi(t(\varepsilon))$ that measures how close the classifier operates to its decision threshold.

4 The Population Effective-Support Floor

The central result prices the robust-optimal discriminant $w^\star(\varepsilon)$—the operating point being certified: its marginal price is bounded below by its own effective support, $\lambda_\varepsilon \geq c_M\sqrt{k_\varepsilon}/\sigma$. (This is a statement about the priced operating point, not a minimax over all discriminants.)

Theorem 4.1 (effective-support floor of the adversarial shadow price; population, with margin-density condition)**.** *Consider the standard setting $x = y\mu + z$, $z \sim N(0, \sigma^2 I_d)$, and let $w^\star(\varepsilon)$ be the robust-optimal linear discriminant. Assume:* (nondegeneracy) *$\varepsilon < \varepsilon_{\text{active}} := \max_i|\mu_i|$, so the active support $S_\varepsilon = \{i : |\mu_i| > \varepsilon\}$ is nonempty, $(|\mu| - \varepsilon)_+ \neq 0$, and k_ε is well-defined; and* (margin-density) *the standardized operating margin satisfies $|t(\varepsilon)| \leq M$ for a constant M. Then*

$$\lambda_\varepsilon = \frac{\|w^\star\|_1}{\sigma\|w^\star\|_2}\,\phi\big(t(\varepsilon)\big) \;\geq\; c_M\,\frac{\sqrt{k_\varepsilon}}{\sigma}, \qquad c_M = \phi(M), \quad k_\varepsilon = \Big(\tfrac{\|w^\star\|_1}{\|w^\star\|_2}\Big)^2,$$

where the constant c_M depends on M (hence on the SNR ρ) only, and involves neither the sample size nor a proportional-asymptotic ratio; the proportional-asymptotic sharp constant is given in Proposition 5.1. For a delocalized dense signal, $k_\varepsilon = \Theta(d)$ and the bound recovers the $\Omega(\sqrt{d})$ special case. As $\varepsilon \uparrow \varepsilon_{\text{active}}$ the active support shrinks to the top coordinate and the robust Bayes advantage vanishes, $\sup_w[\frac{1}{2} - R(\varepsilon; w)] \to 0$; the soft-threshold minimizer degenerates ($w^\star \to 0$ as a limit, not attained—$w = 0$ is not a valid discriminant, $\operatorname{sign}\langle 0, x\rangle$ being undefined), so no floor is claimed for $\varepsilon \geq \varepsilon_{\text{active}}$, consistently with the two-threshold picture of Remark 4.3. (Proof in Appendix B.2.)

Where k_ε comes from: a three-step soft-threshold derivation. The definition of k_ε is not imposed; it is a direct consequence of the KKT/soft-threshold structure (the full argument is in Appendix B.2).

(1) Since Φ is monotone, minimizing the robust risk $R(\varepsilon; w) = \Phi(t(w))$ is equivalent to minimizing $\varepsilon\|w\|_1 - \langle w, \mu\rangle$ subject to $\|w\|_2 = 1$.

(2) Aligning signs (set $u_i = \operatorname{sign}(\mu_i)\, w_i$, so that the optimum has $u \geq 0$), the problem becomes the sphere-positive-orthant linear program

$$\max_{\|u\|_2=1,\ u\geq 0}\ \sum_i \big(|\mu_i| - \varepsilon\big)\, u_i.$$

(3) The optimal direction of this linear functional is $u \propto (|\mu| - \varepsilon)_+$, i.e. $w_i^\star \propto \operatorname{sign}(\mu_i)(|\mu_i| - \varepsilon)_+$.

Consequently the effective support $k_\varepsilon = (\|w^\star\|_1/\|w^\star\|_2)^2$ is the participation ratio of the soft-thresholded solution: it is what KKT/soft-thresholding produce, not an external definition. This is the step that turns $\sqrt{k_\varepsilon}$ from a guessed pricing quantity into a derived one. The threshold is exactly ε because the ℓ_∞ attack term $\varepsilon\|w\|_1 = \varepsilon\sum_i|w_i|$ competes coordinate-wise, through its subgradient $\varepsilon\operatorname{sign}(w_i)$, with the signal μ_i: coordinate i stays active iff $|\mu_i| > \varepsilon$.

Remark 4.2 (margin-density: operational meaning). The window $|t(\varepsilon)| \leq M$ is not a technical device but the operational statement that the classifier operates at a nontrivial robust-accuracy point—the regime in which robustness genuinely costs something. Outside it, no floor is needed: (i) as $t \to -\infty$ (i.e. $\varepsilon \to 0$, high confidence/high SNR) the classifier is far from threshold, robust accuracy $\to 1$, and the marginal cost decays like $\phi(t) \to 0$ (an over-confident classifier pays almost nothing at the margin for extra robustness); (ii) as $t \to +\infty$ (ε past the certification radius $\varepsilon_{\text{cert}}$) robust accuracy $\to 0$ and the linear discriminant carries no information, so neither the analysis nor the floor applies. The $\sqrt{k_\varepsilon}$ law describes precisely the regime in which robustness has a nontrivial price.

Remark 4.3 (two thresholds: $\varepsilon_{\text{active}}$ vs. $\varepsilon_{\text{cert}}$). The support window is governed by $\varepsilon_{\text{active}}$, the active-support collapse scale $\approx \max_i|\mu_i|$ at which the soft-threshold $(|\mu| - \varepsilon)_+$ starts losing $\Theta(d)$ active coordinates. This is a geometric quantity, and it is conceptually distinct from the certification radius $\varepsilon_{\text{cert}} := \langle w^\star, \mu\rangle/\|w^\star\|_1$ (the radius at which $R = \frac{1}{2}$): for a delocalized dense signal the two share the order $\Theta(\|\mu\|_2/\sqrt{d})$ but are not in general equal.

Corollary 4.4 (sparse signal)**.** *Replace assumption (i) of Theorem 4.1 by a k-sparse mean μ supported on k coordinates* whose effective signal $a_\varepsilon = (|\mu| - \varepsilon)_+$ is delocalized on its support *(participation ratio $\Theta(k)$; a spiky μ instead gives $k_\varepsilon = O(1)$, the same subtlety as Lemma 6.2). Then $k_\varepsilon = \Theta(k)$ and $\lambda_\varepsilon \geq c_M \sqrt{k}/\sigma$.* (Proof in Appendix B.4.)

5 Finite Samples and Distributional Universality

The population floor of Theorem 4.1 carries an absolute constant. In the proportional-asymptotic regime the sharp constant is pinned down by the CGMT; and the Gaussian-mixture assumption can be relaxed to subgaussian for a fixed discriminant.

Proposition 5.1 (proportional-asymptotic sharp constant: a conditional corollary of the CGMT characterization)**.** *Place the dense setting of Theorem 4.1 in the normalized Gaussian mixture $x \mid y \sim N(y\mu, I_d)$ with $\|\mu\|_2 = 1$ (a general σ is a rescaling), $\mu_i \overset{iid}{\sim} D$, and take the discriminant to be the ridge-regularized robust ERM*

$$\hat{\theta}(\varepsilon) := \arg\min_{\theta} \sum_{i=1}^{m} L\Big(y_i\langle x_i, \theta\rangle - \tfrac{\varepsilon_{\mathrm{tr}}}{\sqrt{d}}\|\theta\|_1\Big) + r\|\theta\|_2^2,$$

with L convex decreasing and $r > 0$ (the main-theorem radius maps to their scaling via $\varepsilon = \varepsilon_{\mathrm{tr}}/\sqrt{d}$, so the per-coordinate threshold matches the signal scale $|\mu_i| = \Theta(1/\sqrt{d})$ under $\|\mu\|_2 = 1$), in the proportional regime $m, d \to \infty$, $\delta := m/d \to \kappa \in (0, \infty)$. Suppose the data are not $(\ell_\infty, \varepsilon)$-separable, so that the eight-scalar convex–concave saddle-point problem of Taheri, Pedarsani & Thrampoulidis (2022, ISIT, Thm. 1, Eq. 7)—in variables $(\alpha, \tau_1, \omega, \nu, \tau_2, \beta, \gamma, \eta)$ (we write $\omega^\star$ for their scalar order parameter, reserving $w^\star$ for our robust-optimal vector), built from the Moreau envelopes of L and of $|\cdot|$ (the ℓ_1 dual of the ℓ_∞ attack)—has a unique bounded solution, with components $(\omega^\star, \nu^\star, \alpha^\star)$. Then their Lemma 2 gives the order-parameter limits

$$\varepsilon_{\mathrm{tr}}\|\hat{\theta}\|_1/\sqrt{d} \to \omega^\star, \quad \|P_\mu\hat{\theta}\|_2 \to \nu^\star, \quad \|P_\mu^\perp\hat{\theta}\|_2 \to \alpha^\star, \ \text{so } \|\hat{\theta}\|_2 \to \sqrt{\nu^{\star 2} + \alpha^{\star 2}},$$

where P_μ projects onto μ. Taking the ratio yields the exact *readout*

$$\sqrt{k_\varepsilon(\hat{\theta})} = \frac{\|\hat{\theta}\|_1}{\|\hat{\theta}\|_2} = c(\kappa)\sqrt{d} + o(\sqrt{d}), \qquad c(\kappa) = \frac{\omega^\star}{\varepsilon_{\mathrm{tr}}\sqrt{\nu^{\star 2} + \alpha^{\star 2}}} > 0,$$

so $k_\varepsilon(\hat{\theta}) = c(\kappa)^2 d = \Theta(d)$. Substituting into Lemma 3.1, the proportional-asymptotic robust price is $\hat{\lambda}_\varepsilon = (c(\kappa)\sqrt{d}/\sigma)\,\phi(t)$ in the margin-density window, sharpening the population constant $c(\rho)$ of Theorem 4.1 to the explicit, κ-dependent $c(\kappa)$. As $\kappa \to \infty$ we expect *$c(\kappa) \to c(\rho)$ (the sum of m loss terms dominates the fixed ridge, so the effective regularization vanishes and the saddle point formally reduces to the ridgeless population optimum); a rigorous limit requires uniform control of the fixed point as $\kappa \to \infty$, which we do not carry out here. It is consistent with the adversarial error approaching the Bayes-robust error as $\delta \to \infty$ in their analysis.* (Derivation in Appendix B.3.)

This is a *conditional corollary of an explicit external theorem*: it invokes the precise CGMT characterization of Taheri et al. (2022) (isotropic case) and Javanmard & Soltanolkotabi (2022) (anisotropic case) and reads off the single scalar $\|\hat{\theta}\|_1/\|\hat{\theta}\|_2$; we prove no new CGMT result. Its validity is conditional on the non-separability and the uniqueness of their fixed point. The non-Gaussian universality question is Conjecture 5.7.

Remark 5.2 (assumption cross-check with the cited CGMT results). This proposition does not rebuild the CGMT machinery; it aligns our setting with two published precise analyses and specializes. (i) Data model: our binary Gaussian mixture $x = y\mu + z$, $z \sim N(0, \sigma^2 I)$, is the isotropic special case of the general anisotropic mixture of Javanmard & Soltanolkotabi (2022) and the correlated-feature mixture of Taheri et al. (2022) (the anisotropic Σ of Section 6 lies within the former). (ii) Perturbation: our ℓ_∞ attack is covered by their ℓ_p/ℓ_q $(p, q \geq 1)$ analyses. (iii) Loss: our convex non-increasing surrogate matches their "suitable decreasing convex loss" / convex ERM. (iv) Regularization: our ridge is covered by their extension to general convex regularization, and its strong convexity gives uniqueness of $\hat{\theta}$ and of the fixed point. (v) Regime: the proportional asymptotics $n/d \to \kappa$ matches. (vi) Read-out: their characterizations track a small set of scalar

Our object	CGMT order parameter	Source
$\varepsilon_{\mathrm{tr}}\|\hat\theta\|_1/\sqrt{d}$	$\omega^\star$ (scalar ℓ_1-order parameter)	TPT22, Lemma 2
$\|P_\mu\hat\theta\|_2$ (signal overlap)	$\nu^\star$	TPT22, Lemma 2
$\|P_\mu^\perp\hat\theta\|_2$ (noise component)	$\alpha^\star$	TPT22, Lemma 2
saddle point in $(\alpha,\tau_1,\omega,\nu,\tau_2,\beta,\gamma,\eta)$	unique fixed point	TPT22, Thm 1, Eq. 7
readout $\|\hat\theta\|_1/\|\hat\theta\|_2 = c(\kappa)\sqrt{d}$	$c(\kappa)=\omega^\star/\big(\varepsilon_{\mathrm{tr}}\sqrt{\nu^{\star 2}+\alpha^{\star 2}}\big)$	this paper (Prop. 5.1)

Table 2: Mapping for Proposition 5.1: each object we use is a published CGMT order parameter (Taheri et al., 2022, "TPT22"); the last row is the single scalar we read off. We prove no new CGMT result.

order parameters (including $\|\hat\theta\|_2$, the correlation with μ, and the ℓ_1-type quantity produced by the ℓ_∞ dual); we merely read off the single ratio $\|\hat\theta\|_1/\|\hat\theta\|_2$ and prove its $\sqrt{d}$ scaling in the dense limit—this specialization is our contribution, not a new CGMT result.

Proposition 5.3 (subgaussian universality for a fixed dense discriminant)**.** *Fix a discriminant w with $\|w\|_\infty/\|w\|_2 = O(1/\sqrt{d})$, and let the noise have independent, zero-mean, subgaussian coordinates (parameter $\tau^2>0$) that additionally admit a bounded density or satisfy Cramér's condition, so that a local central limit theorem applies. Then the robust marginal price $\lambda_\varepsilon(w)=\|w\|_1\, p_S\big(\varepsilon\|w\|_1-\langle w,\mu\rangle\big)$ retains its $\sqrt{d}$ prefactor, the constant entering through p_S and τ. The local-CLT hypothesis is necessary: Berry–Esseen controls only the CDF F_S, whereas the price uses the density p_S, whose convergence requires a Gnedenko-type local limit theorem.* (Proof in Appendix B.5.) *Established (partial: fixed discriminant).*

5.1 Beyond Gaussian: a distribution-free floor for symmetric log-concave noise

The Gaussian assumption in Theorem 4.1 enters only through the density factor $\phi(t)\ge\phi(M)$. Log-concavity supplies the same bound for free, which removes the Gaussian assumption for the core floor.

Lemma 5.4 (log-concave density floor)**.** *Let g be a symmetric log-concave density on $\mathbb{R}$ with variance* 1 *and CDF F. Then for every level $\alpha_0\in(0,\frac12]$ there is a constant $c(\alpha_0)>0$,* independent of *g, with $g(t)\ge c(\alpha_0)$ whenever $F(t)\ge\alpha_0$ and $t\le 0$; the explicit value $c(\alpha_0)=\alpha_0/4$ is admissible (via $g(0)=\|g\|_\infty\ge\frac18$ and $g(t)\ge 2g(0)F(t)$). Consequently, if z has iid symmetric log-concave coordinates (variance σ^2), then for* every *w the standardized projection $\tilde S=\langle w,z\rangle/(\sigma\|w\|_2)$ is symmetric log-concave with unit variance, and at a margin-density operating point (robust accuracy $F(t)\ge\alpha_0$) its density obeys $p_{\tilde S}(t)\ge c_M:=c(\alpha_0)$. The bound is absolute (independent of the noise law) and, being tied to the operating CDF level, needs no support caveat.* (Proof in Appendix B.6.)

Theorem 5.5 (distribution-free effective-support floor; symmetric log-concave noise)**.** *Let z have iid symmetric log-concave coordinates with variance σ^2 (Gaussian, Laplace, logistic, hyperbolic-secant, uniform, ...). At the robust-optimal discriminant $w^\star(\varepsilon)$ operating in the margin-density window $|t(w^\star)|\le M$,*

$$\lambda_\varepsilon(w^\star)=\frac{\sqrt{k_\varepsilon}}{\sigma}\,p_{\tilde S^*}\big(t(w^\star)\big)\;\ge\;c_M\,\frac{\sqrt{k_\varepsilon}}{\sigma},$$

with c_M the absolute density floor of Lemma 5.4, independent of the noise law within the symmetric log-concave class*. This extends the effective-support floor of Theorem 4.1 from Gaussian to all symmetric log-concave noise, using no CLT and no proportional-asymptotic limit. It is a k_ε-*floor*: it prices $w^\star$ by its own effective support and does* not *by itself yield $\Omega(\sqrt{d})$—the dimensional statement requires the delocalization $k_\varepsilon=\Theta(d)$ of Proposition 5.6.* (Proof in Appendix B.7.)

The floor above prices $w^\star$ by its own k_ε. To recover the dense $\Omega(\sqrt{d})$ statement one must further know that $w^\star$ is delocalized ($k_\varepsilon=\Theta(d)$)—which, unlike the floor, is genuinely distribution-dependent (the argmin need not equal the Gaussian soft-threshold). The next result gives a clean, checkable sufficient condition.

Proposition 5.6 (distribution-free dense floor; a checkable delocalization condition)**.** *Let $a_\varepsilon=(|\mu|-\varepsilon)_+$ be the effective signal, $\hat a_\varepsilon=a_\varepsilon/\|a_\varepsilon\|_2$ its dense soft-threshold direction with participation fraction $c:=$*

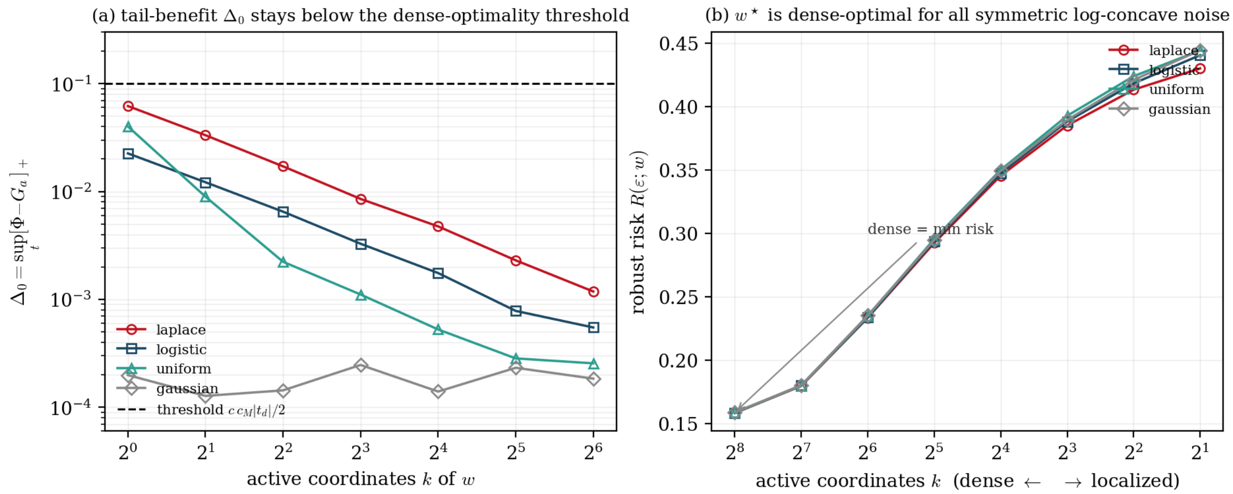

Figure 1: Distributional universality of the effective-support floor. **(a)** *A numerical certificate of the hypothesis of Proposition 5.6*: the maximal Gaussian undershoot $\Delta_0 = \sup_t [\Phi - G_a]_+$ (the tail-benefit of localizing) versus the number of active coordinates k, for Laplace/logistic/uniform noise; it is maximal at $k = 1$, *decreases along convolution*, and stays below the dense-optimality threshold $\frac{1}{2} c\, \phi_M |t_d|$ (dashed, $\phi_M = \phi(M)$) for all three at the stated operating point—verifying the checkable inequality (Gaussian gives $\Delta_0 \approx 0$). **(b)** *An empirical illustration of the conclusion*: the robust risk $R(\varepsilon; w)$ as w is localized (soft-threshold restricted to its top-k coordinates) is minimized at the dense end ($k = d$) for every symmetric log-concave law, consistent with $w^\star$ being dense-optimal ($k_\varepsilon = \Theta(d)$). Panel (a) certifies the sufficient condition at this operating point (not for all operating points or all log-concave laws); panel (b) illustrates the resulting dense-optimality.

$k_\varepsilon(a_\varepsilon)/d \in (0, 1]$, and $t_d = -\|a_\varepsilon\|_2/\sigma = \min_w t(w)$ the optimal standardized margin, and $\phi_M := \phi(M)$ the Gaussian density at the window edge (distinct from the log-concave floor c_M of Lemma 5.4). Define the maximal Gaussian CDF-undershoot *of the standardized noise sum,*

$$\Delta_0 := \sup_{\|a\|_2=1} \sup_t \left[\Phi(t) - G_a(t)\right]_+, \qquad G_a := \mathrm{law}\Big(\textstyle\sum_i a_i \zeta_i\Big).$$

Intuitively, localizing w buys a thinner left tail (worth at most Δ_0 in risk) but costs standardized margin (worth ϕ_M per unit deficit); the dense soft-threshold wins whenever the margin cost dominates. If $\Delta_0 < \frac{1}{2} c\, \phi_M\, |t_d|$, then the robust-optimal $w^\star$ is delocalized, with

$$k_\varepsilon(w^\star) \;\geq\; c\, d \left(1 - \sqrt{\tfrac{2\Delta_0}{\phi_M |t_d|}}\right)^2 = \Theta(d), \qquad \textit{hence} \qquad \lambda_\varepsilon \;\geq\; c_M \frac{\sqrt{d}}{\sigma}.$$

For Gaussian noise $\Delta_0 = 0$, so the condition holds unconditionally (recovering the dense case of Theorem 4.1); for Laplace, logistic, and uniform noise it is verified at moderate operating points (Figure 1). Status: *the implication "condition ⇒ dense floor" is fully proven (Appendix B.8); what is numerical is the* verification of the hypothesis $\Delta_0 < \frac{1}{2} c\, \phi_M |t_d|$ *for the non-Gaussian laws (Figure 1a). Thus the dense $\Omega(\sqrt{d})$ floor is Established for Gaussian and* Established conditional on the checkable inequality*—numerically certified, not proven in general—for symmetric log-concave noise.* (Proof in Appendix B.8.)

Conjecture 5.7 (unconditional distribution-free floor)**.** *The dense floor of Proposition 5.6 holds unconditionally for all symmetric log-concave noise—equivalently $\Delta_0 \leq \delta_1 := \sup_t[\Phi(t) - F_0(t)]_+$, the single-coordinate undershoot, which Figure 1(a) shows is small and is the convolution-worst case—and, beyond log-concavity, for any noise with mild moment and anti-concentration conditions. The obstruction is that the Kolmogorov-distance undershoot is not monotone along convolution in general, so $\Delta_0 \leq \delta_1$ is not automatic; a direct bound, or the full non-log-concave case, is open.*

6 A Conditional Stable-Rank Specialization

This section is *not* a general replacement of k_ε by the stable rank $r_{\text{eff}} = \operatorname{tr}(\Sigma)/\|\Sigma\|$: the effective support k_ε remains the pricing quantity throughout. Rather, Proposition 6.3 gives a stack of conditions (EO0–EO4) under which k_ε *specializes* to $\Theta(r_{\text{eff}})$—i.e. under which the weight-space effective support collapses onto the data-space stable rank. Outside those conditions k_ε and r_{eff} can differ (Remark 6.6).

Lemma 6.1 (deterministic prefactor interval)**.** *Fix the ℓ_∞ attack basis and, throughout this section, assume* (EO0) *that Σ is diagonal in it (so the ℓ_1/ℓ_2 geometry and the Σ-eigenbasis are aligned; the non-aligned case carries a coherence correction, Remark 6.6). Then for any w the geometric prefactor $P(w) := \|w\|_1/\|w\|_\Sigma$ (with $\|w\|_\Sigma^2 := w^\top \Sigma w$) obeys the deterministic bounds*

$$\frac{1}{\sqrt{\|\Sigma\|}} \cdot \frac{\|w\|_1}{\|w\|_2} \;\le\; P(w) \;\le\; \frac{1}{\sqrt{\lambda_{\min}(\Sigma)}} \cdot \frac{\|w\|_1}{\|w\|_2},$$

so the price factor is squeezed between the flat-spectrum value ($P = \frac{\|w\|_1}{\|w\|_2}/\sqrt{\|\Sigma\|}$, attained on the top-eigenvalue subspace) and the inverse-covariance-inflated value (attained when w aligns with the smallest-variance directions, where $P \to \sqrt{\operatorname{tr}(\Sigma^{-1})}$). (Proof in Appendix B.11.)

Lemma 6.2 (flat-spectrum exact value)**.** *If Σ restricted to a subspace V of dimension r is exactly flat ($\Sigma|_V = \|\Sigma\|\, I_r$) and w is supported on V, then the* exact identity $P(w) = \frac{1}{\sqrt{\|\Sigma\|}} \cdot \frac{\|w\|_1}{\|w\|_2}$ *holds. In particular* $P(w) = \sqrt{r}/\sqrt{\|\Sigma\|}$ if and only if *w is delocalized on V, i.e. its participation ratio $(\|w\|_1/\|w\|_2)^2 = \Theta(r)$; for a spiky w (e.g. one coordinate) $P(w)$ can be as small as $1/\sqrt{\|\Sigma\|}$. Support on V alone does not give the $\sqrt{r}$ factor.* (Proof in Appendix B.12.)

Proposition 6.3 ($k_\varepsilon = \Theta(r_{\text{eff}})$ under aligned effective occupancy)**.** *Let $r_{\text{eff}} := \operatorname{tr}(\Sigma)/\|\Sigma\| \in [1, d]$, let V_{eff} be the span of the top $\lceil r_{\text{eff}} \rceil$ eigendirections of Σ, and write the* effective signal *$a_\varepsilon := (|\mu| - \varepsilon)_+$ (in the aligned case below the robust-optimal weight is $w_i^\star \propto \operatorname{sign}(\mu_i)\, a_{\varepsilon,i}/\lambda_i$, so it is a_ε, not μ, that sets $w^\star$'s participation ratio). Fix the ℓ_∞ attack basis and assume:* (EO0) basis alignment—*$\Sigma = \operatorname{diag}(\lambda_i)$ in the attack basis, so V_{eff} is spanned by attack-basis coordinates and the ℓ_1 geometry is aligned with the eigenbasis;* (EO1) sub-spectrum flatness *$\lambda_{\max}(\Sigma|_{V_{\text{eff}}})/\lambda_{\min}(\Sigma|_{V_{\text{eff}}}) \le 1+\rho'$;* (EO2) signal-tail control *(in the Σ^{-1} metric) $\|P_{V_{\text{eff}}^\perp}\mu\|_{\Sigma^{-1}}^2 \le \beta\, \|P_{V_{\text{eff}}}\mu\|_{\Sigma^{-1}}^2$, with $\|v\|_{\Sigma^{-1}}^2 := v^\top \Sigma^{-1} v$;* (EO3) effective-signal delocalization—*a_ε restricted to V_{eff} is delocalized, $\max_{i \in V_{\text{eff}}} a_{\varepsilon,i}/\|P_{V_{\text{eff}}} a_\varepsilon\|_2 = O(1/\sqrt{r_{\text{eff}}})$, so its participation ratio is $\Theta(r_{\text{eff}})$; and* (EO4) subgradient-tail control—*the ℓ_1 subgradient $s \in \partial\|w^\star\|_1$ obeys $\|P_{V_{\text{eff}}^\perp}\Sigma^{-1}(\varepsilon s)\|_\Sigma \le \rho''\, \|P_{V_{\text{eff}}} w^\star\|_\Sigma$ (the attack term does not pump mass into the low-variance tail). Then, combining (EO2) and (EO4), the tail of $w^\star$ obeys*

$$\|P_{V_{\text{eff}}^\perp} w^\star\|_\Sigma \;\le\; O(\beta + \rho' + \rho'')\, \|P_{V_{\text{eff}}} w^\star\|_\Sigma,$$

so the Mahalanobis principal mass of $w^\star$ concentrates on V_{eff} (tail leakage controlled, not exact concentration); by (EO1) $\|w^\star\|_\Sigma^2 = (1 + O(\rho'))\|\Sigma\|\|P_{V_{\text{eff}}} w^\star\|_2^2$, and by (EO3) the participation ratio of $w^\star|_{V_{\text{eff}}}$ (proportional to $a_\varepsilon|_{V_{\text{eff}}}$) is $\Theta(r_{\text{eff}})$, so

$$P(w^\star) \ge (1 - O(\rho' + \beta + \rho''))\, \frac{\sqrt{r_{\text{eff}}}}{\sqrt{\|\Sigma\|}},$$

with equality in the fully aligned, flat, occupied, delocalized limit (Lemma 6.2); hence $\lambda_\varepsilon \ge c\, \frac{\sqrt{r_{\text{eff}}}}{\sqrt{\|\Sigma\|}}\, p_S(\varepsilon\|w^\star\|_1 - \langle w^\star, \mu\rangle)$. Three failure modes: large β (signal leaks into the low-variance tail, $P(w^\star) \to \sqrt{\operatorname{tr}\Sigma^{-1}} \gg \sqrt{r_{\text{eff}}}$); large ρ'' (subgradient pumps tail mass); and violated (EO3) (spiky $a_\varepsilon|_{V_{\text{eff}}}$, whose solution sits on $O(1)$ coordinates, $P(w^\star) = O(1)$). (Proof in Appendix B.13.) *Conditional; region of validity delimited by (ρ', β, ρ'') and the delocalization (EO3). This gives the non-adaptive, basis-aligned $\sqrt{r_{\text{eff}}}$ half of Conjecture 6.5; the non-aligned case is Remark 6.6.*

Remark 6.4 ((EO4) is an assumption, not a consequence of (EO2))**.** (EO2) and (EO4) control different tails and are logically independent. (EO2) bounds the *signal* tail $\Sigma^{-1}\mu$; (EO4) separately rules out *adversarial subgradient-tail pumping*—the ℓ_1 subgradient $s \in \partial\|w^\star\|_1$ of the attack driving $\Sigma^{-1}s$ mass into low-variance directions. Because s is pinned down only up to the subdifferential (any $s_i \in [-1, 1]$ on inactive coordinates), (EO4) cannot be derived from (EO2); it is a genuine hypothesis, to be posited or verified per instance. It holds trivially in the degenerate construction of Figure 2, where the tail carries neither signal nor active coordinates (so the subgradient tail is empty); it is *not* implied by small tail eigenvalues—those make Σ^{-1} large on the tail, i.e. the regime in which subgradient-tail pumping is most dangerous—and is otherwise a real assumption.

Conjecture 6.5 (adaptive-feature refinement). *Under adaptive feature learning, where the representation—and hence Σ—is shaped by training, the $\sqrt{r_{\text{eff}}}$ floor holds with r_{eff} the stable rank of the* learned *representation. The rigorous adaptive version (beyond the non-adaptive Proposition 6.3) is open.*

Remark 6.6 (non-aligned covariance and coherence). Assumption (EO0) is essential: $k_\varepsilon = (\|\cdot\|_1/\|\cdot\|_2)^2$ is measured in the ℓ_∞ attack basis, whereas r_{eff} and V_{eff} are spectral. When Σ is not diagonal in the attack basis, the identification $P(w^\star) \approx \sqrt{r_{\text{eff}}/\|\Sigma\|}$ acquires a mutual-coherence factor $\gamma := \max_i \|P_{V_{\text{eff}}} e_i\|_2$ between the attack coordinates e_i and the top eigensubspace, which we do not control quantitatively (we state no coherence-corrected bound). A fully coherent Σ (eigenvectors delocalized in the attack basis) can decouple k_ε from r_{eff} entirely—the pricing dimension then remains k_ε, but it no longer collapses onto r_{eff}. Establishing a sharp coherence-corrected floor in the non-aligned case is open.

7 The Local Deep-Network Extension

The same $\sqrt{k_\varepsilon}$ prefactor appears locally: at the operating point of any C^2 classifier, the first-order sensitivity carries the effective-support prefactor of its linearized discriminant. This is a *local* statement (a first-order proxy at a working point), not a global floor for the network.

Proposition 7.1 (local first-order sensitivity; deterministic worst case). *Let $f\colon \mathbb{R}^d \to \mathbb{R}$ be a C^2 classifier, x_0 a working point, $w := \nabla f(x_0)$, $y := \operatorname{sign}(f(x_0))$, and let $\kappa_\infty(x_0) := \sup_{s\in\{\pm1\}^d} |s^\top \nabla^2 f(x_0)\, s|$ be the ℓ_∞ quadratic-form curvature (a natural curvature measure on the ℓ_∞ ball, avoiding the d factor of a spectral-norm bound). For an ℓ_∞-bounded adversary $\|\delta\|_\infty \le \varepsilon$, define the local adversarial margin $g(\varepsilon; x_0) := \min_{\|\delta\|_\infty\le\varepsilon} y\, f(x_0+\delta)$. Then, on the first-order-valid neighborhood $\varepsilon^2\kappa_\infty(x_0) = o(\varepsilon\|w\|_1)$,*

$$g(\varepsilon; x_0) = y\, f(x_0) - \varepsilon\|w\|_1 + O(\varepsilon^2\kappa_\infty), \qquad \left.\frac{\partial g}{\partial \varepsilon}\right|_{\varepsilon\to 0} = -\|\nabla f(x_0)\|_1,$$

(deterministic, worst-case, distribution-free). Applying Theorem 4.1 to the linearized discriminant $w = \nabla f(x_0)$: under independent input noise providing a density p_S and the margin-density condition, the local robust marginal price is pinned by the prefactor $\|\nabla f(x_0)\|_1/\|\nabla f(x_0)\|_2 = \sqrt{k_\varepsilon}$—$\Theta(\sqrt{d})$ for dense ∇f, $\sqrt{r_{\text{eff}}}$ under low effective rank (Proposition 6.3). (Proof in Appendix B.9.) *Established (the worst-case margin sensitivity is deterministic; the density part of the $\sqrt{d}$ price factor inherits the random-perturbation and margin-density conditions of Theorem 4.1).*

Remark 7.2 (scope: a local proxy, not a global deep-net theorem). Proposition 7.1 is a first-order (linearized) proxy at the working point x_0: it applies Theorem 4.1 to the local linearization $w = \nabla f(x_0)$, and its conclusion is confined to the first-order-valid neighborhood, bounded by the κ_∞ remainder. It is *not* a global robustness theorem for deep networks; a rigorous global lower bound across many layers of nonlinearity is an explicitly open problem.

Remark 7.3 (estimating κ_∞ for ReLU networks). For a ReLU network, $\nabla^2 f$ is a.e. zero (piecewise linear) and the curvature concentrates, as a singular measure, on the activation-boundary hyperplanes; the classical Hessian spectrum does not apply. Near x_0 one should read κ_∞ as an effective curvature estimated by the gradient jump across the nearest activation boundaries, $\kappa_\infty(x_0) \approx \sum_j \|\nabla f(x_0^+) - \nabla f(x_0^-)\|/\operatorname{dist}(x_0, H_j)$ over boundaries H_j in the ε-neighborhood (nearer boundaries contributing more). One forward pass yields the activation pattern, the boundary distances, and the two-sided gradient difference, making this a practical proxy for the second-order correction.

Corollary 7.4 (differentiable surrogate loss). *For a differentiable surrogate loss and the smoothed classifier it induces, the local first-order sensitivity of the surrogate robust risk is bounded below by $c\, m_0\, \|\nabla f(x_0)\|_1/\|\nabla f(x_0)\|_2 = c\, m_0\sqrt{k_\varepsilon}$, where $m_0 > 0$ is a local derivative lower bound of the surrogate at the operating margin.* (Proof in Appendix B.10.)

8 Implications and Validation

8.1 A cheap proxy and the measurement bridge

As $\varepsilon \to 0$ the expensive robust-side price is estimated by a cheap training-side quantity.

Proposition 8.1 ($\varepsilon \to 0$ cheap proxy)**.** *As $\varepsilon \to 0$, the robust marginal price is estimated by the training-side input-gradient dual norm $\hat{\lambda}_\varepsilon := \mathbb{E}\|\nabla_x \ell(x, y)\|_1$, obtainable from a single backward pass.* (Proof in Appendix A.5.)

Proposition 8.2 (Danskin first-order expansion; the Hessian term)**.** *Assume $\ell(\cdot, y)$ is C^2 in an ε-neighborhood of x with L_3-Lipschitz Hessian. Then*

$$|\lambda_\varepsilon - \hat{\lambda}_\varepsilon| \le \varepsilon\, \mathbb{E}[\kappa_\infty] + O(L_3 \varepsilon^2),$$

so the proxy is exact to first order, with the correction controlled by the ℓ_∞ quadratic-form curvature κ_∞. (Proof in Appendix A.6.)

Remark 8.3 (computability of κ_∞). Computing $\kappa_\infty(x) = \max_{s \in \{\pm 1\}^d} s^\top \nabla^2 f(x)\, s$ exactly is a Boolean quadratic program and NP-hard, but this does not impede the engineering bridge. First, κ_∞ only enters the price correction $\varepsilon\, \mathbb{E}[\kappa_\infty]$ of Proposition 8.2 (an $O(\varepsilon)$ term that vanishes as $\varepsilon \to 0$), while the measured leading term is $\hat{\lambda}_\varepsilon = \mathbb{E}\|\nabla_x \ell\|_1$ (one backward pass); a computable *upper* bound on κ_∞ suffices, and the crude spectral bound $\kappa_\infty \le \sqrt{d}\, \|\nabla^2 f\|_F$ (Hutchinson trace estimation) is computable, but since κ_∞ enters the $O(\varepsilon)$ price term, its $\sqrt{d}$ factor makes that correction $O(\varepsilon\sqrt{d})$—usable only at very small ε; the dimension-free SDP upper bound below is preferable. Second, κ_∞ admits a computable dimension-free *upper* bound via a semidefinite relaxation of the Boolean quadratic program (in the PSD case the $\pi/2$-theorem (Nesterov, 1998) gives a $2/\pi$ approximation ratio; in the indefinite case the SDP still returns a valid upper bound, though we do not claim a sharp two-sided approximation ratio), while random sign-vector sampling gives a lower bound. The upper bound is all the engineering bridge needs, since κ_∞ enters only the $O(\varepsilon)$ correction term.

8.2 The certification radius

Corollary 8.4 (certification radius; binary-channel reduction)**.** *For the linear discriminant $w^\star$, the certification radius at which the robust risk reaches $\frac{1}{2}$ is $\varepsilon_{\text{cert}} = \langle w^\star, \mu\rangle / \|w^\star\|_1$; beyond it the classifier is worse than chance and, under the fixed binary-channel reduction, carries no information.* (Proof in Appendix B.14.) *Established (within that reduction).*

8.3 The optimal robust radius

Proposition 8.5 (optimal robust radius ε^{**}; existence and comparative statics)**.** *Let a deployment value function $V : [0, \infty) \to \mathbb{R}_+$ and a robustness cost $C : [0, \infty) \to \mathbb{R}_+$ satisfy:* (a) $V(\varepsilon) = K(1 - P_{\text{breach}}(\varepsilon))$ *with P_{breach} strictly decreasing, C^1, $P_{\text{breach}}(0) = 1$, $P_{\text{breach}}(\infty) = 0$, and $K > 0$ the single-breach loss;* (b) $C(\varepsilon) = \int_0^\varepsilon \lambda_\varepsilon(\varepsilon')\, d\varepsilon'$*, with marginal cost $C'(\varepsilon) = \lambda_\varepsilon$ the robust price at ε (Theorem 4.1 lower-bounds it by* $c_M\sqrt{k_\varepsilon}/\sigma$ inside *the margin-density window), and $\lambda_0 := C'(0^+) = \lim_{\varepsilon \to 0^+} \lambda_\varepsilon$ the marginal price at $\varepsilon \to 0^+$ (the over-confident regime), generally $\lambda_0 \ll c_M\sqrt{k_\varepsilon}/\sigma$;* (c) $V'(\varepsilon) = K|P'_{\text{breach}}(\varepsilon)|$ *strictly decreasing. Let $\varepsilon^{**} := \arg\max_\varepsilon [V(\varepsilon) - C(\varepsilon)]$. Then:* (i) *an interior optimum $\varepsilon^{**} > 0$ exists iff $K|P'_{\text{breach}}(0)| > C'(0^+) = \lambda_0$ (net marginal benefit positive at 0; note the criterion is the true marginal cost $C'(0^+) = \lambda_0$, not the margin-density window bound $c_M\sqrt{k_\varepsilon}/\sigma$);* (ii) *$\varepsilon^{**}$ is unique;* (iii) *the boundary condition is $K|P'_{\text{breach}}(\varepsilon^{**})| = \lambda_\varepsilon(\varepsilon^{**})$;* (iv) *comparative statics through an* exogenous *complexity parameter π (e.g. the ambient dimension d, or r_{eff} as a fixed data/representation property) that shifts the whole cost schedule upward, $\partial\lambda_\varepsilon/\partial\pi > 0$: then $\partial\varepsilon^{**}/\partial\pi < 0$ (raising marginal cost lowers the optimal radius), and in particular $\partial\varepsilon^{**}/\partial K > 0$, $\partial\varepsilon^{**}/\partial d < 0$ (a "high-complexity model is economically hard to robustify"). We differentiate through the exogenous π,* not *through the endogenous $k_\varepsilon = k_\varepsilon(\varepsilon)$, which cannot be held fixed while varying the choice variable ε;* (v) *under the fixed binary-channel reduction of Corollary 8.4, $\varepsilon^{**} \le \varepsilon_{\text{cert}}$.* (Proof in Appendix C.) *Conditional; P_{breach} is supplied by the deployment threat model, not first principles.*

Remark 8.6 (reduced-form model). Proposition 8.5 is a reduced-form economic model, not an attacker–defender game: $P_{\text{breach}}(\varepsilon)$ is the breach probability against a fixed budget-bounded ℓ_∞ threat model, decreasing in the certified radius by construction. An adaptive attacker only reshapes P_{breach} (its worst-case

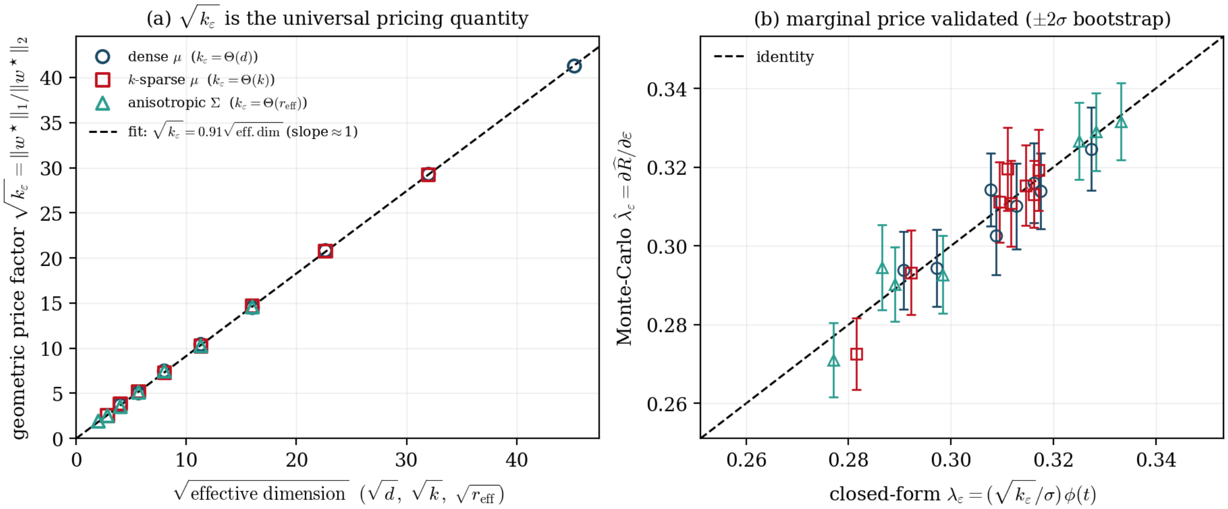

Figure 2: Synthetic validation (d, k, r_{eff} swept; $\sigma \propto \|\mu\|_2$ at constant SNR $\rho = 1.5$, operating point $t \approx -1$ in the margin-density window). **(a)** The geometric price factor $\sqrt{k_\varepsilon} = \|w^\star\|_1/\|w^\star\|_2$ for dense μ, k-sparse μ, and flat-spectrum anisotropic Σ (satisfying effective occupancy) collapses onto a single line against $\sqrt{\text{effective dimension}}$ (linear slope ≈ 1, i.e. $\sqrt{k_\varepsilon} = 0.91\sqrt{\text{eff. dim}}$; equivalently a log–log dimensional-tax exponent $\approx \frac{1}{2}$), with $\sqrt{d}$, $\sqrt{k}$, $\sqrt{r_{\mathrm{eff}}}$ its three special cases. **(b)** The Monte-Carlo marginal price $\partial\widehat{R}/\partial\varepsilon$ matches the closed form $(\sqrt{k_\varepsilon}/\sigma)\phi(t)$ on the identity line; error bars are $\pm 2\sigma$ (bootstrap, 100 resamples; relative $\leq 3.7\%$).

over strategies stays monotone in the radius while the budget is bounded), so the first-order condition and comparative statics—which need only monotone P_{breach} and convex cost—survive. The full game is a natural extension.

8.4 Synthetic validation

Figure 2 reports a synthetic experiment validating the main result. Across three regimes—dense μ, k-sparse μ, and a flat-subspace anisotropic Σ satisfying effective occupancy—the geometric price factor $\sqrt{k_\varepsilon} = \|w^\star\|_1/\|w^\star\|_2$ collapses onto a single line against $\sqrt{\text{effective dimension}}$: a linear fit gives $\sqrt{k_\varepsilon} = 0.91\sqrt{\text{eff. dim}}$ (slope ≈ 1); equivalently, the log–log slope of $\sqrt{k_\varepsilon}$ versus the effective dimension is $\approx \frac{1}{2}$—the "dimensional-tax exponent" $\frac{1}{2}$, the log–log form of the same $\sqrt{\cdot}$-scaling law (the two are two coordinate readings of one relation, not two different slopes). The coefficient $0.91 = \sqrt{0.836}$ is the participation-ratio constant of the soft-threshold at the operating ε ($k_\varepsilon/(\text{eff. dim}) \approx 0.836 < 1$), an operating-point constant rather than a deviation from the $\sqrt{\cdot}$-law. Thus $\sqrt{d}$, $\sqrt{k}$, and $\sqrt{r_{\mathrm{eff}}}$ are the three special cases $k_\varepsilon = \Theta(d), \Theta(k), \Theta(r_{\mathrm{eff}})$ (Theorem 4.1, Corollary 4.4, Proposition 6.3). The Monte-Carlo marginal price $\partial\widehat{R}/\partial\varepsilon$ agrees with the closed form $(\sqrt{k_\varepsilon}/\sigma)\phi(t(\varepsilon))$ on the identity line, validating the closed-form risk and price of Lemma 3.1; the operating point takes $\sigma \propto \|\mu\|_2$ (constant SNR $\rho = 1.5$), placing $t \approx -1$ inside the margin-density window (so $c_M = \phi(-1) \approx 0.242$, matching the fitted price constant).

Technical details. Signal coordinates $\mu_i \sim \mathrm{Unif}(0.5, 1.5)$ (on the support for the dense/sparse regimes, on the flat top-r subspace with vanishing tail eigenvalues for the anisotropic regime); dimensions swept over $d \in \{16, \ldots, 2048\}$ (dense), $k \in \{8, \ldots, 1024\}$ (sparse, $d = 2048$), $r \in \{4, \ldots, 256\}$ (anisotropic, $d = 512$); the working point ε is fixed within the active regime, and $\sigma \propto \|\mu\|_2$ (constant SNR $\rho = 1.5$) places $t \approx -1$ in the margin-density window; $w^\star(\varepsilon)$ is the closed-form soft-threshold; the Monte Carlo uses $N = 2 \times 10^6$ samples, sampling the projection $\langle w, z\rangle \sim N(0, \|w\|_\Sigma^2)$ directly, with $\hat{\lambda}_\varepsilon$ estimated by central differences; the slope is a least-squares fit through the origin; the seed is fixed for reproducibility.

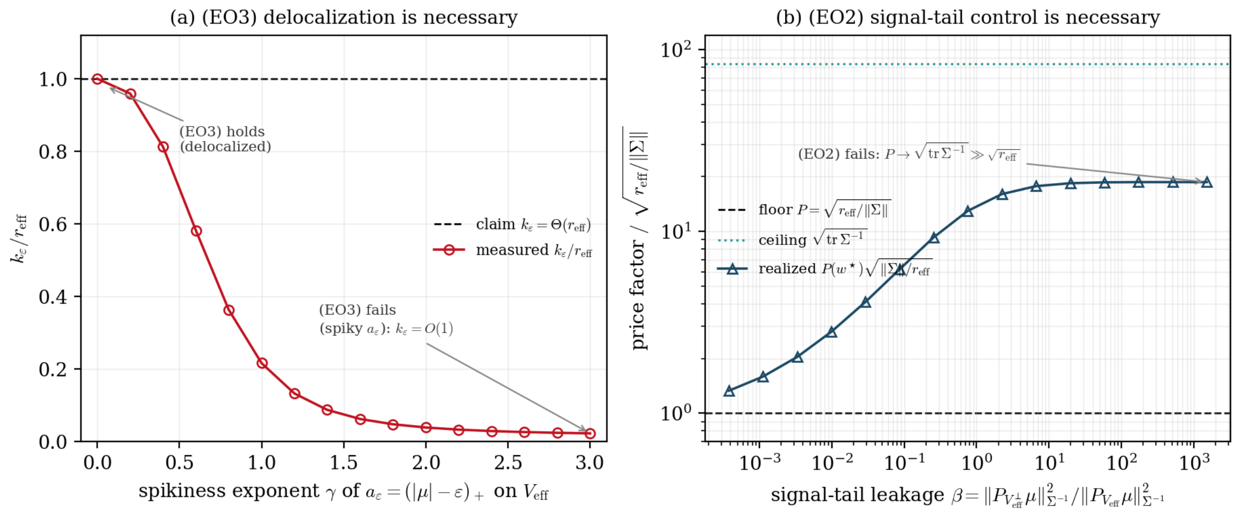

Figure 3: The stable-rank specialization is conditional ($d = 512$, flat top-r_{eff}=64 subspace). **(a)** Violating (EO3): as the effective signal $a_\varepsilon = (|\mu| - \varepsilon)_+$ is made spiky (increasing exponent γ), $k_\varepsilon/r_{\mathrm{eff}}$ falls from 1 (delocalized) toward $O(1)/r_{\mathrm{eff}}$—support on V_{eff} alone does not give $\Theta(r_{\mathrm{eff}})$. **(b)** Violating (EO2): as signal leaks into the low-variance tail (increasing β), the realized price factor $P(w^\star)\sqrt{\|\Sigma\|/r_{\mathrm{eff}}}$ rises from 1 (the floor) toward the $\sqrt{\mathrm{tr}\,\Sigma^{-1}}$ ceiling. The conditions are necessary, not decorative.

The occupancy conditions are necessary (Figure 3). Two sweeps confirm that the conditions of Proposition 6.3 are not decorative. Violating (EO3) by making the effective signal a_ε spiky on the flat subspace drives the measured $k_\varepsilon/r_{\mathrm{eff}}$ from 1 down toward $O(1)/r_{\mathrm{eff}}$ (from 1.00 to 0.02 over the sweep). Violating (EO2) by leaking signal into the low-variance tail (raising β) drives the realized price factor above the $\sqrt{r_{\mathrm{eff}}/\|\Sigma\|}$ floor toward the $\sqrt{\mathrm{tr}\,\Sigma^{-1}}$ ceiling (here up to ≈19× the floor). Outside (EO0)–(EO4), k_ε does not collapse onto r_{eff}.

8.5 Effective rank and low-rank adaptation (discussion)

This subsection is discussion, not a consequence of the theorems. The $\sqrt{r_{\mathrm{eff}}}$ specialization of Proposition 6.3 suggests a resource-economic lens: if a learned representation had stable rank $r_{\mathrm{eff}} \ll d$ and satisfied the aligned conditions (EO0)–(EO4), the dimensional tax on robustness would relax from $\sqrt{d}$ toward $\sqrt{r_{\mathrm{eff}}}$. This gives a conditional reading of why low-rank adaptation (LoRA/PEFT) can maintain robust behavior with few degrees of freedom—consistent with the price being set by r_{eff} rather than the ambient dimension—though we claim no causal or quantitative link, and the rigorous adaptive-feature version is the open Conjecture 6.5. Order-of-magnitude (illustrative only): since the price scales as $\sqrt{r_{\mathrm{eff}}}$, a drop from $r_{\mathrm{eff}} \approx 10^4$ to $\approx 10^2$ would lower the prefactor by $\sqrt{10^4/10^2} = 10$.

9 Related Work and Limitations

9.1 Related work

Positioning. This paper relates to, but is distinct from, five lines of work. **(a) Shadow prices / sensitivity analysis** (Fiacco, 1983; Rockafellar & Wets, 1998; Danskin, 1966; Clarke, 1990) provides the general envelope/subgradient theory of constrained extrema; we specialize it to the adversarial radius as a one-dimensional budget and identify its pricing unit $\sqrt{k_\varepsilon}$. **(b) CGMT + adversarial training** (Thrampoulidis et al., 2015; Taheri et al., 2022; Javanmard & Soltanolkotabi, 2022) precisely characterizes the test error/accuracy of adversarial training; our departure is to *read off* $\|\hat{w}\|_1/\|\hat{w}\|_2 = \sqrt{k_\varepsilon}$ from the order parameters as a pricing quantity and connect it to a geometric floor. **(c) Structured/low-rank covariance**

and adversarial robustness: recent high-dimensional analyses tie robustness to the feature-covariance structure—Javanmard & Soltanolkotabi (2022) for anisotropic Gaussian mixtures, and Tanner et al. (2024), whose Block Feature Model captures feature-dependent robustness under structured (positive-definite) attacks and defenses; we supply the *pricing mechanism* (the $\sqrt{r_{\text{eff}}}$ refinement, Proposition 6.3, conditional), connecting low effective rank to a shadow price. **(d) Participation ratio as an effective dimension**: the participation ratio and its reciprocal (the inverse participation ratio) are standard measures of localization and effective dimension in condensed-matter physics and random-matrix theory (Anderson, 1958; Wegner, 1980; Evers & Mirlin, 2008), and of the effective dimensionality of data in computational neuroscience (Gao et al., 2017); using it as the operational *pricing* quantity $k_\varepsilon = (\|w^\star\|_1/\|w^\star\|_2)^2$ of adversarial robustness is, to our knowledge, new. **(e) $\Omega(\sqrt{d})$-type robustness lower bounds** (Schmidt et al., 2018; Bubeck & Sellke, 2021; Yin et al., 2019) give dimensional lower bounds on robustness cost from a sample-complexity/capacity angle; our floor is about the marginal price (shadow price) itself, and refines $\sqrt{d}$ to the special case $\sqrt{k_\varepsilon}$.

Robust machine learning. Our marginal-price lower bound builds on the robust-learning literature: sample-complexity lower bounds and the robustness–accuracy trade-off (Schmidt et al., 2018; Yin et al., 2019; Tsipras et al., 2019; Bubeck & Sellke, 2021) characterize the cost of robustness from the "how much data/accuracy is needed" angle; we are complementary, casting the cost of robustness as the marginal of a value function in the radius and giving the dimensional lower bound $\Omega(\sqrt{k_\varepsilon})$—which is $\Omega(\sqrt{d})$ in the dense case and $\Omega(\sqrt{r_{\text{eff}}})$ under low effective rank. That the ℓ_∞/ℓ_1 dual geometry saturates in high dimension is a well-known intuition; our contribution is to embed it in the subgradient framework of a value function and to give the population $\sqrt{k_\varepsilon}$ geometric floor (Theorem 4.1, with an absolute constant from the soft-threshold active support), while the proportional-asymptotic sharp constant is supplied by the CGMT (Proposition 5.1).

High-dimensional statistics and the CGMT. The proportional-asymptotic sharp constant $c(\kappa)$ of Proposition 5.1 is pinned down by the convex Gaussian min–max theorem (Thrampoulidis et al., 2015); the precise characterization of $\hat{w}$ under an ℓ_∞ attack, binary classification, and a Gaussian mixture is due to Taheri et al. (2022) and Javanmard & Soltanolkotabi (2022), from which we specialize to read the $\sqrt{k_\varepsilon}$ scaling (the $\sqrt{d}$ scaling in the dense case). The $\sqrt{k_\varepsilon}$ floor of Theorem 4.1 is itself population geometry, independent of proportional asymptotics, with absolute constant $c(\rho)$ from the soft-threshold active support. Related techniques appear in precise trade-offs for robust linear regression (Javanmard et al., 2020) and the effect of regularization in high-dimensional logistic regression (Salehi et al., 2019).

9.2 Limitations

The floor is proved in a Gaussian–linear regime; the distribution-free version (Conjecture 5.7) and the adaptive-feature effective-rank version (Conjecture 6.5) are open. The deep-network extension (Proposition 7.1) is a local first-order proxy at a working point, not a global robustness theorem across many nonlinear layers. The optimal-radius result (Proposition 8.5) is a reduced-form economic model in which the breach probability P_{breach} is supplied by the deployment threat model rather than derived from first principles. Finally, the CGMT step (Proposition 5.1) is a conditional corollary of an external precise-asymptotics theorem, valid under the non-separability and fixed-point uniqueness of the cited results.

10 Discussion and Conclusion

We have argued that the marginal price of adversarial robustness $\lambda_\varepsilon = \partial L^*/\partial\varepsilon$, viewed as the subgradient of a constrained value function, has a dimensional floor whose correct pricing unit is the effective support $k_\varepsilon = (\|w^\star\|_1/\|w^\star\|_2)^2$ of the robust-optimal discriminant. The core statement is that the operational dimension of adversarial pricing is k_ε, not the ambient dimension d: ℓ_∞ attack $\Rightarrow$ ℓ_1 dual geometry $\Rightarrow$ the prefactor $\|w^\star\|_1/\|w^\star\|_2 = \sqrt{k_\varepsilon}$, and in the margin-density window $\lambda_\varepsilon \geq c_M\sqrt{k_\varepsilon}/\sigma$. The ambient dimension $\sqrt{d}$, the sparsity $\sqrt{k}$, and the stable rank $\sqrt{r_{\text{eff}}}$ are three special cases $k_\varepsilon = \Theta(d), \Theta(k), \Theta(r_{\text{eff}})$; the proportional-asymptotic sharp constant is supplied by the CGMT; and the bound transfers to the local operating point of any C^2 classifier through the ℓ_∞ quadratic-form curvature κ_∞. A synthetic experiment confirms that $\sqrt{k_\varepsilon}$ is the universal pricing quantity across all three regimes.

The engineering reading is that the dimensional tax on robustness is dictated by the ℓ_∞/ℓ_1 geometry rather than optional, and that its relaxation from $\sqrt{d}$ to $\sqrt{r_{\mathrm{eff}}}$—conditional on the aligned effective-occupancy conditions (EO0)–(EO4) of Section 6—offers a conditional account of why low-effective-rank representations could lower the marginal cost of robustness and alignment. Two open problems stand out: a distribution-free floor for the optimal discriminant beyond the Gaussian design (Conjecture 5.7), and a rigorous adaptive-feature $\sqrt{r_{\mathrm{eff}}}$ refinement in which the representation, and hence the covariance, is shaped by training (Conjecture 6.5). The latter, in particular, would decide whether $\Omega(\sqrt{d})$ is a universal lower bound or an artifact of the lazy/fixed-feature regime.

A Preparatory Propositions

A.1 Proof of Theorem 2.1 (variational principle)

Under (A1) the problem $\min\{L(x) : g_i(x) \le \mathcal{E}_i\}$ is a convex program; under (A2) (Slater or MFCQ) the KKT conditions are necessary and sufficient for optimality, so there exist multipliers $\lambda \ge 0$ with $\nabla L(x^*) + \sum_i \lambda_i \nabla g_i(x^*) = 0$ and $\lambda_i(g_i(x^*) - \mathcal{E}_i) = 0$ (see, e.g., Rockafellar & Wets, 1998; Bertsekas, 1999). Multipliers need not be unique. Under (A3) LICQ the active-gradient system has full column rank, so the stationarity equation determines λ uniquely; alternatively (A3′) strict complementarity with a second-order sufficient condition yields uniqueness by the implicit-function theorem applied to the KKT system. The differentiability restriction and its equivalence to uniqueness are proved in Appendix A.2. □

A.2 Proof of Proposition 2.2 (shadow price = gradient)

This is the (nonsmooth) envelope theorem. Writing the Lagrangian $\mathcal{L}(x, \lambda; \mathcal{E}) = L(x) + \sum_i \lambda_i(g_i(x) - \mathcal{E}_i)$, at an optimum $\partial\mathcal{L}/\partial\mathcal{E}_i = -\lambda_i$. When L^* is differentiable at $\mathcal{E}$ (equivalently, the multiplier is unique, by (A3)/(A3′)), Danskin's theorem gives $\partial L^*/\partial\mathcal{E}_i = -\lambda_i$. In general L^* is the pointwise minimum of affine-in-$\mathcal{E}$ functions and hence concave and locally Lipschitz, so by Rademacher's theorem it is differentiable a.e.; at nondifferentiable points $-\lambda \in \partial_{\mathcal{E}} L^*$ (the Clarke subdifferential), which is the convex hull of the multiplier set and reduces to a Danskin interval in the ε direction. □

A.3 Proof of Corollary 2.3 (marginal equality)

At the optimum of $\max\{-L^*(\mathcal{E}) : \sum_i \mathcal{E}_i \le B\}$ the stationarity condition equates $\partial L^*/\partial\mathcal{E}_i = -\lambda_i$ across all active coordinates to a common Lagrange multiplier of the total-budget constraint; hence the marginal loss reduction per unit budget is equalized across active dimensions. □

A.4 The readout operator (deferred from Section 2)

Definition A.1 (readout operator)**.** For a closed subspace $Y \subseteq X$ with inclusion $\iota_Y : Y \to X$, the readout of ∂L^* along Y is the inclusion-adjoint restriction $\Pi_Y : X^* \to Y^*$, $\Pi_Y \partial L^* = \{\, \xi|_Y : \xi \in \partial L^* \,\}$. Two distinguished readouts are $\Pi_{\mathrm{train}} := \Pi_U$ (along the training control) and $\Pi_{\mathrm{robust}} := \Pi_\varepsilon$ (along the adversarial radius).

Proposition A.2 (the two readouts)**.** *Under the setting of Section 2 and where L^* is single-valued, $\Pi_{\mathrm{train}}(\nabla L^*) = \lambda(t)$ (the costate) and $\Pi_{\mathrm{robust}}(\partial L^*) = \partial_\varepsilon L^* = [\lambda_\varepsilon^-, \lambda_\varepsilon^+]$ (the Clarke subdifferential, a.e. single-valued, a Danskin interval on the critical set). Consequently the costate readout and the robust readout are the same object seen along different axes—"change the axis, not the machine"—and, as $\varepsilon \to 0$, the robust price equals a training-side input-gradient dual norm up to a Hessian remainder (quantified in Proposition 8.2).*

Proof. The claim is that $\Pi_Y = \iota_Y^*$, the adjoint of the inclusion $\iota_Y : Y \to X$, restricted to ∂L^*. For a subgradient $\xi \in \partial L^* \subseteq X^*$, its action on Y is $\langle \xi, \iota_Y y\rangle = \langle \iota_Y^* \xi, y\rangle$, i.e. $\xi|_Y = \iota_Y^* \xi$. Taking $Y = U$ (training control) recovers the costate $\lambda(t)$ via Pontryagin's principle (the costate is exactly the sensitivity of the value to the control), and $Y = \mathbb{R}_\varepsilon$ recovers $\partial_\varepsilon L^*$. Since all three are images of the single set ∂L^* under restriction, they are readouts of one object; the a.e. single-valuedness follows from Appendix A.2. □

A.5 Proof of Proposition 8.1 ($\varepsilon \to 0$ cheap proxy)

By Danskin's theorem, $\partial_\varepsilon \mathbb{E} \max_{\|\delta\|_\infty \le \varepsilon} \ell(x+\delta, y)\big|_{\varepsilon=0} = \mathbb{E} \max_{\|u\|_\infty \le 1} \langle u, \nabla_x \ell(x,y)\rangle = \mathbb{E}\|\nabla_x \ell(x,y)\|_1$, since the ℓ_∞-ball support function is the ℓ_1 norm (the worst-case unit direction is $u = \mathrm{sign}(\nabla_x \ell)$). Thus $\hat{\lambda}_\varepsilon = \mathbb{E}\|\nabla_x \ell\|_1$ is the exact $\varepsilon \to 0$ derivative, obtained from one backward pass. $\square$

A.6 Proof of Proposition 8.2 (Hessian term)

Let $\delta^*(\varepsilon) = \arg\max_{\|\delta\|_\infty \le \varepsilon} \ell(x+\delta, y)$. A second-order Taylor expansion gives $\ell(x+\delta^*, y) = \ell(x,y) + \langle \nabla_x \ell, \delta^* \rangle + \frac{1}{2} \delta^{*\top} \nabla_x^2 \ell \, \delta^* + O(L_3 \varepsilon^3)$. The leading term maximizes at $\delta^* = \varepsilon \, \mathrm{sign}(\nabla_x \ell)$ giving $\varepsilon \|\nabla_x \ell\|_1$; the quadratic correction is bounded by $\frac{1}{2} \max_{s \in \{\pm 1\}^d} \varepsilon^2 \, s^\top \nabla_x^2 \ell \, s = \frac{\varepsilon^2}{2} \kappa_\infty$. The remainder $R(\varepsilon) := \frac{1}{2} \delta^{*\top} \nabla_x^2 \ell \, \delta^* = \frac{\varepsilon^2}{2} s^\top \nabla_x^2 \ell \, s$ (with $s = \mathrm{sign}(\nabla_x \ell)$) has $|R(\varepsilon)| \le \frac{\varepsilon^2}{2} \kappa_\infty$, but the *price* is its derivative $R'(\varepsilon) = \varepsilon \, s^\top \nabla_x^2 \ell \, s$, so $|R'(\varepsilon)| \le \varepsilon \kappa_\infty$. Taking expectations, $|\lambda_\varepsilon - \hat{\lambda}_\varepsilon| \le \varepsilon \, \mathbb{E}[\kappa_\infty] + O(L_3 \varepsilon^2)$. $\square$

B Main Proofs

B.1 Proof of Lemma 3.1 (closed-form robust risk and price)

For $x = y\mu + z$ and a linear discriminant, the (signed) clean margin is $y\langle w, x\rangle = \langle w, \mu \rangle + y\langle w, z\rangle$. Against an ℓ_∞ adversary of radius ε, the worst-case perturbation solves $\min_{\|\delta\|_\infty \le \varepsilon} y\langle w, x+\delta\rangle = y\langle w, x\rangle - \varepsilon\|w\|_1$ (the ℓ_∞-ball support function of w is $\varepsilon\|w\|_1$), so misclassification occurs iff $y\langle w, x\rangle < \varepsilon\|w\|_1$. Since $y\langle w, z\rangle \sim N(0, \sigma^2\|w\|_2^2)$ (by symmetry of y and z),

$$R(\varepsilon; w) = \mathbb{P}\big(\langle w, \mu\rangle + y\langle w, z\rangle < \varepsilon\|w\|_1\big) = \Phi\left(\frac{\varepsilon\|w\|_1 - \langle w, \mu\rangle}{\sigma\|w\|_2}\right) = \Phi(t(w)).$$

Differentiating with w fixed (Danskin: at the optimum the total derivative equals the partial), $\partial R / \partial \varepsilon = \phi(t) \cdot \|w\|_1 / (\sigma \|w\|_2) = (\sqrt{k_\varepsilon(w)}/\sigma)\phi(t)$, with $k_\varepsilon(w) = (\|w\|_1/\|w\|_2)^2$. $\square$

B.2 Proof of Theorem 4.1 (effective-support floor)

Step 1 (reduction). By Lemma 3.1, $R(\varepsilon; w) = \Phi(t(w))$ with $t(w) = (\varepsilon\|w\|_1 - \langle w, \mu\rangle)/(\sigma\|w\|_2)$. Since Φ is strictly increasing, the robust-optimal $w^\star(\varepsilon) = \arg\min_w R = \arg\min_w t(w)$, and by scale invariance we may impose $\|w\|_2 = 1$; the problem becomes $\min_{\|w\|_2 = 1} \ \varepsilon\|w\|_1 - \langle w, \mu\rangle$.

Step 2 (sign alignment). At the optimum, $\mathrm{sign}(w_i) = \mathrm{sign}(\mu_i)$ for every active coordinate (flipping a sign strictly decreases $-\langle w, \mu\rangle$ without changing $\|w\|_1, \|w\|_2$). Writing $u_i = \mathrm{sign}(\mu_i) w_i \ge 0$, the problem is $\max_{\|u\|_2 = 1,\, u \ge 0} \ \sum_i (|\mu_i| - \varepsilon) u_i$.

Step 3 (soft-threshold solution). This is the maximization of a linear functional over the intersection of the unit sphere with the positive orthant; the optimal direction is the positive part of the coefficient vector, $u \propto (|\mu| - \varepsilon)_+$, i.e.

$$w_i^\star \propto \mathrm{sign}(\mu_i)\,(|\mu_i| - \varepsilon)_+.$$

Only coordinates with $|\mu_i| > \varepsilon$ survive, because the ℓ_∞ attack term $\varepsilon\|w\|_1$ contributes a per-coordinate subgradient $\varepsilon \ \mathrm{sign}(w_i)$ that competes with the signal μ_i; the active support is $S_\varepsilon = \{i : |\mu_i| > \varepsilon\}$.

Step 4 (floor). By construction $k_\varepsilon = (\|w^\star\|_1/\|w^\star\|_2)^2$ is the participation ratio of $(|\mu| - \varepsilon)_+$ over S_ε. By Lemma 3.1, $\lambda_\varepsilon = (\sqrt{k_\varepsilon}/\sigma)\phi(t(\varepsilon))$. On the margin-density window $|t(\varepsilon)| \le M$, the density is bounded below, $\phi(t) \ge \phi(M) =: c_M$ (using that ϕ is unimodal and even, so $\phi(t) \ge \phi(M)$ for $|t| \le M$), whence $\lambda_\varepsilon \ge c_M \sqrt{k_\varepsilon}/\sigma$. For a delocalized dense signal ($|\mu_i| = \Theta(\|\mu\|_2/\sqrt{d})$), $|S_\varepsilon| = \Theta(d)$ and the participation ratio of an ℓ_∞-comparable vector is $\Theta(d)$, giving $k_\varepsilon = \Theta(d)$ and the $\Omega(\sqrt{d})$ special case. The constant $c_M = \phi(M)$ depends on M (hence on the operating SNR ρ) only. $\square$

B.3 Derivation of Proposition 5.1 (explicit readout from the CGMT fixed point)

This is a specialization of an external theorem, not a new CGMT proof. *Reduction.* For a decreasing loss the inner ℓ_∞ maximization is solved by $\delta_i^* = -y_i \operatorname{sign}(\theta)\,\varepsilon_{\rm tr}/\sqrt{d}$, reducing the robust ERM to the convex program $\min_\theta \sum_i L\big(y_i\langle x_i,\theta\rangle - \frac{\varepsilon_{\rm tr}}{\sqrt{d}}\|\theta\|_1\big) + r\|\theta\|_2^2$ (Taheri et al., 2022, Eq. 9), in which the ℓ_∞ attack has become an "effective" ℓ_1 term inside the loss. *Scalarization.* Their CGMT analysis (their Thm. 1) scalarizes the associated auxiliary optimization to the eight-scalar convex–concave saddle point (their Eq. 7) in $(\alpha,\tau_1,\omega,\nu,\tau_2,\beta,\gamma,\eta)$ (ω their scalar order parameter, distinct from our vector $w^\star$), whose Moreau envelope of $|\cdot|$ encodes the ℓ_∞ attack through its ℓ_1 dual; under non-separability and ridge strong convexity the solution is unique. *Readout.* Their Lemma 2 identifies the order-parameter limits $\varepsilon_{\rm tr}\|\hat\theta\|_1/\sqrt{d} \to \omega^\star$, $\|P_\mu\hat\theta\|_2 \to \nu^\star$, $\|P_\mu^\perp\hat\theta\|_2 \to \alpha^\star$, so $\|\hat\theta\|_2 \to \sqrt{\nu^{\star 2}+\alpha^{\star 2}}$. Taking the ratio,

$$\frac{\|\hat\theta\|_1}{\|\hat\theta\|_2} = \frac{\omega^\star/\varepsilon_{\rm tr}}{\sqrt{\nu^{\star 2}+\alpha^{\star 2}}}\sqrt{d} + o(\sqrt{d}) = c(\kappa)\sqrt{d}, \qquad c(\kappa) = \frac{\omega^\star}{\varepsilon_{\rm tr}\sqrt{\nu^{\star 2}+\alpha^{\star 2}}}.$$

For dense D (delocalized μ) the fixed point has $\omega^\star > 0$ and $\sqrt{\nu^{\star 2}+\alpha^{\star 2}} = \Theta(1)$, so $\|\hat\theta\|_1 = \Theta(\sqrt{d})$, $\|\hat\theta\|_2 = \Theta(1)$, and $k_\varepsilon(\hat\theta) = c(\kappa)^2 d = \Theta(d)$. Substituting into Lemma 3.1 gives $\hat\lambda_\varepsilon = (c(\kappa)\sqrt{d}/\sigma)\phi(t)$ in the margin-density window. *Population limit.* As $\kappa = m/d \to \infty$ the sum of m loss terms dominates the fixed ridge, the effective regularization $\to 0$, and the saddle point of Eq. 7 formally reduces to the unregularized population program of Theorem 4.1, so we *expect* $c(\kappa) \to c(\rho)$; we do not carry out the uniform fixed-point control that a rigorous limit would require. The assumption cross-check is Remark 5.2. □

B.4 Proof of Corollary 4.4 (sparse signal)

If μ is supported on k coordinates, then $S_\varepsilon = \{i : |\mu_i| > \varepsilon\} \subseteq \operatorname{supp}(\mu)$ has at most k elements, and for ε below the sparse-signal scale $|S_\varepsilon| = \Theta(k)$. By Step 3–4 of Appendix B.2, $w_i^\star \propto \operatorname{sign}(\mu_i)a_{\varepsilon,i}$ is supported on S_ε, so $k_\varepsilon = (\|w^\star\|_1/\|w^\star\|_2)^2$ is the participation ratio of a_ε on S_ε; this is $\Theta(k)$ *iff a_ε is delocalized there* (support alone is insufficient, cf. Lemma 6.2). Under that delocalization, $k_\varepsilon = \Theta(k)$ and $\lambda_\varepsilon \geq c_M\sqrt{k}/\sigma$. □

B.5 Proof of Proposition 5.3 (subgaussian universality)

Fix w delocalized ($\|w\|_\infty/\|w\|_2 = O(1/\sqrt{d})$). The projection $S := \langle w, z\rangle = \sum_i w_i z_i$ is a sum of independent zero-mean subgaussian terms with $\max_i|w_i|/\|w\|_2 \to 0$, so by a Lindeberg/local CLT argument the standardized $S/\|w\|_2$ converges to $N(0,\tau^2)$; the delocalization and the local-CLT hypothesis (bounded density / Cramér's condition) give convergence of the *density* p_S, not merely the CDF. Then $\lambda_\varepsilon(w) = \|w\|_1\, p_S(\varepsilon\|w\|_1 - \langle w,\mu\rangle)$ retains the $\|w\|_1 = \Theta(\sqrt{d}\,\|w\|_2)$ prefactor, with the constant carried by p_S and τ. Berry–Esseen alone bounds $|F_S - \Phi|$ but not $|p_S - \phi|$, which is why the local-limit hypothesis is required. □

B.6 Proof of Lemma 5.4 (log-concave density floor)

Let g be symmetric log-concave with variance 1 and CDF F; we prove the explicit floor $c_M = \alpha_0/4$ with α_0 the margin-density level (below), by a self-contained reverse-hazard argument.

Step 1 ($g(0) \geq \frac{1}{8}$). By Chebyshev $P(|X| \leq 2) \geq 1 - \frac{1}{4} = \frac{3}{4}$, so $F(2) \geq \frac{7}{8}$ and the point q with $F(q) = \frac{3}{4}$ obeys $q \leq 2$. Since symmetry and unimodality make g nonincreasing on $[0,\infty)$, $\frac{1}{4} = \int_0^q g \leq q\,g(0) \leq 2g(0)$, hence $g(0) = \|g\|_\infty \geq \frac{1}{8}$.

Step 2 (reverse hazard). Log-concavity of g makes F log-concave, so the reverse hazard $g/F = (\log F)'$ is nonincreasing (Bagnoli & Bergstrom, 2005); see also the survey of Saumard & Wellner (2014). Thus for every $t \leq 0$, $g(t)/F(t) \geq g(0)/F(0) = 2g(0)$, i.e. $g(t) \geq 2g(0)F(t)$.

Step 3 (floor at the operating point). The margin-density window is precisely the statement that the robust accuracy is nontrivial: the operating point has $F(t(w^\star)) = R(\varepsilon; w^\star) \geq \alpha_0 > 0$ for a constant α_0 (equivalently $|t| \leq M$ with $\alpha_0 = \Phi(-M)$), and $t(w^\star) < 0$ since $R < \frac{1}{2}$. Combining Steps 1–2, $g(t(w^\star)) \geq 2g(0)\alpha_0 \geq$

$\alpha_0/4 =: c_M > 0$, an absolute constant depending only on α_0 (hence on M), independent of the noise law. This also disposes of bounded-support laws (e.g. uniform): any operating point with $F(t) \geq \alpha_0 > 0$ is interior to the support, where $g > 0$, so no support caveat is needed beyond the margin-density hypothesis itself.

Finally, for iid symmetric log-concave z_i, the projection $\langle w, z\rangle$ is a convolution of symmetric log-concave laws, hence symmetric log-concave (Prékopa's theorem); after normalizing to unit variance, $\tilde{S}$ satisfies the hypotheses and $p_{\tilde{S}}(t) \geq c_M$ on the window. □

B.7 Proof of Theorem 5.5 (distribution-free floor)

The price identity of Lemma 3.1 holds for any noise with a density: $\lambda_\varepsilon(w^\star) = \|w^\star\|_1\, p_S(\tau) = \frac{\|w^\star\|_1}{\sigma\|w^\star\|_2} p_{\tilde{S}^*}(t(w^\star)) = \frac{\sqrt{k_\varepsilon}}{\sigma} p_{\tilde{S}^*}(t(w^\star))$, where $\tilde{S}^* = \langle w^\star, z\rangle/(\sigma\|w^\star\|_2)$. By Lemma 5.4, $\tilde{S}^*$ is symmetric log-concave with unit variance, so at the margin-density operating point $|t(w^\star)| \leq M$ (interior to the support, since the robust accuracy is nontrivial there) its density satisfies $p_{\tilde{S}^*}(t(w^\star)) \geq c_M$. Therefore $\lambda_\varepsilon(w^\star) \geq c_M\sqrt{k_\varepsilon}/\sigma$, with c_M absolute over the symmetric log-concave class. No CLT or proportional-asymptotic limit is used. □

B.8 Proof of Proposition 5.6 (distribution-free dense floor)

Intuition. The dense soft-threshold minimizes the standardized margin t; a localized competitor can only lower risk by exploiting a thinner-than-Gaussian tail, worth at most Δ_0, while paying a margin deficit worth ϕ_M per unit. Balancing the two gives the threshold $\Delta_0 < \frac{1}{2}c\,\phi_M|t_d|$. Formally: align signs so $\mu_i > 0$, $w_i \geq 0$; on active coordinates $t(w) = -\langle\hat{w}, a_\varepsilon\rangle/\sigma$ with $\hat{w} = w/\|w\|_2$ unit. Throughout, $\phi_M := \phi(M)$ denotes the *Gaussian* density at the window edge (used only to lower-bound differences of Φ), distinct from the log-concave density floor c_M of Lemma 5.4 (used only for the final price).

Step 1 (margin-deficit identity). By Cauchy–Schwarz $\min_w t(w) = t_d = -\|a_\varepsilon\|_2/\sigma$, attained at $\hat{w} = \hat{a}_\varepsilon$; for any unit $\hat{w}$,

$$t(w) - t_d = \frac{\|a_\varepsilon\|_2}{\sigma}\big(1 - \langle\hat{w}, \hat{a}_\varepsilon\rangle\big) = \frac{|t_d|}{2}\,\|\hat{w} - \hat{a}_\varepsilon\|_2^2, \tag{†}$$

using $1 - \langle\hat{w}, \hat{a}_\varepsilon\rangle = \frac{1}{2}\|\hat{w} - \hat{a}_\varepsilon\|_2^2$ for unit vectors.

Step 2 (risk comparison). Let $w^\star_{\mathrm{ST}}$ be the dense soft-threshold. Two facts bracket the competitor's risk against it:

(a) *Competitor lower bound.* By the very definition of $\Delta_0 = \sup_{a,t}[\Phi(t) - G_a(t)]_+$, every direction and point obey $G_a(t) \geq \Phi(t) - \Delta_0$; hence $R(w) = G_{\hat{w}}(t(w)) \geq \Phi(t(w)) - \Delta_0$.

(b) *Dense upper bound.* For the delocalized $\hat{a}_\varepsilon$ ($\eta_0 = \|\hat{a}_\varepsilon\|_\infty = O(1/\sqrt{d})$), Berry–Esseen gives $R(w^\star_{\mathrm{ST}}) = G_{\hat{a}_\varepsilon}(t_d) \leq \Phi(t_d) + C_{BE}\rho_3\eta_0 = \Phi(t_d) + o(1)$.

Subtracting (b) from (a) and bounding the Φ-difference by its Gaussian density on the window,

$$R(w) - R(w^\star_{\mathrm{ST}}) \;\geq\; \underbrace{\big[\Phi(t(w)) - \Phi(t_d)\big]}_{\geq\,\phi_M\,(t(w)-t_d)\text{ by MVT, }\phi\geq\phi(M)} - \Delta_0 - o(1) \;\geq\; \phi_M\,[t(w) - t_d] - \Delta_0 - o(1),$$

valid for $t(w) \leq M$ (if $t(w) > M$ then $R(w) \geq \Phi(M) - \Delta_0 > R(w^\star_{\mathrm{ST}})$ trivially, so such w are already excluded).

Step 3 (the Δ_0 threshold). The true optimum has $R(w^\star) \leq R(w^\star_{\mathrm{ST}})$, so the display forces $\phi_M[t(w^\star) - t_d] \leq \Delta_0 + o(1)$; substituting (†),

$$\|\hat{w}^\star - \hat{a}_\varepsilon\|_2^2 \;\leq\; \frac{2\Delta_0}{\phi_M\,|t_d|} + o(1) \;=:\; \theta_0.$$

Thus a nonzero tail-benefit Δ_0 buys the optimizer only a bounded angular deviation θ_0 from the dense direction, and $\theta_0 < 1$ exactly when $\Delta_0 < \frac{1}{2}\phi_M|t_d|$.

Step 4 (delocalization $\Rightarrow k_\varepsilon$). Since $\hat{a}_\varepsilon$ is supported on the cd active coordinates, $\|\hat{w^\star}\|_1 \geq \|\hat{a}_\varepsilon\|_1 - \sqrt{cd}\,\|\hat{w^\star} - \hat{a}_\varepsilon\|_2 \geq \sqrt{cd}\,(1-\sqrt{\theta_0})$, so $k_\varepsilon(w^\star) = \|\hat{w^\star}\|_1^2 \geq cd(1-\sqrt{\theta_0})^2 = \Theta(d)$ provided $\theta_0 < c$, i.e. $\Delta_0 < \frac{1}{2}\,c\,\phi_M\,|t_d|$. Combining with the k_ε-floor of Theorem 5.5 (which supplies the density factor c_M), $\lambda_\varepsilon \geq c_M\sqrt{d}/\sigma$. For Gaussian noise $G_a = \Phi$, so $\Delta_0 = 0$ and the condition is vacuous, recovering the dense case of Theorem 4.1. $\square$

B.9 Proof of Proposition 7.1 (local deep-net sensitivity)

Taylor-expand f at x_0: $f(x_0+\delta) = f(x_0) + \langle \nabla f(x_0), \delta\rangle + \frac{1}{2}\delta^\top \nabla^2 f(x_0)\delta + o(\|\delta\|^2)$. The worst-case ℓ_∞ perturbation minimizing yf takes the first-order part to $-\varepsilon\|\nabla f(x_0)\|_1$ (support function of the ℓ_∞ ball), while the quadratic part is bounded by $\frac{1}{2}\varepsilon^2 \max_{s\in\{\pm1\}^d}|s^\top\nabla^2 f\, s| = \frac{1}{2}\varepsilon^2\kappa_\infty$. Hence $g(\varepsilon;x_0) = yf(x_0) - \varepsilon\|w\|_1 + O(\varepsilon^2\kappa_\infty)$ and $\partial g/\partial\varepsilon \to -\|\nabla f(x_0)\|_1$ as $\varepsilon \to 0$. Treating $w = \nabla f(x_0)$ as the linearized discriminant and applying Lemma 3.1/Theorem 4.1 under input noise with density p_S and margin-density, the local price is bounded below by the prefactor $\|\nabla f(x_0)\|_1/\|\nabla f(x_0)\|_2 = \sqrt{k_\varepsilon}$. Validity is confined to the neighborhood $\varepsilon^2\kappa_\infty = o(\varepsilon\|w\|_1)$ (Remark 7.2). $\square$

B.10 Proof of Corollary 7.4 (differentiable surrogate)

For a differentiable surrogate ψ with $\psi' \leq -m_0 < 0$ at the operating margin, the surrogate robust risk $\mathbb{E}\psi(g(\varepsilon;x_0))$ has ε-derivative $\mathbb{E}[\psi'(g)\,\partial_\varepsilon g]$; using $\partial_\varepsilon g \to -\|\nabla f\|_1$ and $|\psi'| \geq m_0$ gives a lower bound $c\,m_0\|\nabla f(x_0)\|_1/\|\nabla f(x_0)\|_2 = c\,m_0\sqrt{k_\varepsilon}$. $\square$

B.11 Proof of Lemma 6.1 (deterministic interval)

By (EO0), $\Sigma = \mathrm{diag}(\lambda_i)$ in the attack basis. Write $P(w) = \|w\|_1/\|w\|_\Sigma$. Since $\lambda_{\min}(\Sigma)\|w\|_2^2 \leq \|w\|_\Sigma^2 \leq \|\Sigma\|\|w\|_2^2$, we get $\frac{1}{\sqrt{\|\Sigma\|}}\frac{\|w\|_1}{\|w\|_2} \leq P(w) \leq \frac{1}{\sqrt{\lambda_{\min}(\Sigma)}}\frac{\|w\|_1}{\|w\|_2}$. The lower value $\|w\|_1/(\|w\|_2\sqrt{\|\Sigma\|})$ is the flat-spectrum value (top-eigenvalue subspace); the upper value is attained when w aligns with the smallest-variance directions, where $P \to \sqrt{\mathrm{tr}(\Sigma^{-1})}$. Without (EO0), $\|w\|_1$ (attack basis) and $\|w\|_\Sigma$ (eigenbasis) are related only through the coherence factor of Remark 6.6. $\square$

B.12 Proof of Lemma 6.2 (flat spectrum)

If $\Sigma|_V = \|\Sigma\| I_r$ and w is supported on V, then $\|w\|_\Sigma^2 = \|\Sigma\|\|w\|_2^2$, giving the *exact identity* $P(w) = \|w\|_1/(\sqrt{\|\Sigma\|}\|w\|_2) = (\|w\|_1/\|w\|_2)/\sqrt{\|\Sigma\|}$. The factor $\|w\|_1/\|w\|_2 \in [1,\sqrt{r}]$ equals $\sqrt{(\|w\|_1/\|w\|_2)^2}$, the square root of the participation ratio, so $P(w) = \sqrt{r}/\sqrt{\|\Sigma\|}$ holds *iff* w is delocalized on V (participation ratio $\Theta(r)$); a spiky w (e.g. one coordinate) gives $\|w\|_1/\|w\|_2 = 1$ and $P(w) = 1/\sqrt{\|\Sigma\|}$. Support on V alone is insufficient; delocalization is required (and is supplied, for the soft-threshold solution, by (EO3) in Proposition 6.3). $\square$

B.13 Proof of Proposition 6.3 ($\sqrt{r_{\mathrm{eff}}}$ floor)

Work in the ℓ_∞ attack basis, where by (EO0) $\Sigma = \mathrm{diag}(\lambda_i)$; the robust-optimal weight is $w^\star \propto \Sigma^{-1}(\mu - \varepsilon s)$, $s \in \partial\|w^\star\|_1$, $|s_i| \leq 1$, so on active coordinates $w_i^\star \propto \mathrm{sign}(\mu_i)(|\mu_i| - \varepsilon)/\lambda_i = \mathrm{sign}(\mu_i)a_{\varepsilon,i}/\lambda_i$. *Signal mass.* By (EO2), $\|P_{V_{\mathrm{eff}}^\perp}\mu\|_{\Sigma^{-1}}^2 \leq \beta\|P_{V_{\mathrm{eff}}}\mu\|_{\Sigma^{-1}}^2$, so a $(1-O(\beta))$ fraction of the Mahalanobis mass of $\Sigma^{-1}\mu$ lies in V_{eff}, ruling out signal amplification in the low-variance tail. *Subgradient mass.* The ℓ_1-subgradient term contributes tail energy $\|P_{V_{\mathrm{eff}}^\perp}\Sigma^{-1}(\varepsilon s)\|_\Sigma$, which is exactly what (EO4) bounds by $\rho''\|P_{V_{\mathrm{eff}}}w^\star\|_\Sigma$; we take this as a hypothesis rather than deriving it, since s is only pinned down up to the subdifferential. Combining the two tails, $\|P_{V_{\mathrm{eff}}^\perp}w^\star\|_\Sigma \leq O(\beta+\rho'+\rho'')\|P_{V_{\mathrm{eff}}}w^\star\|_\Sigma$. *Prefactor.* By (EO1), $\|w^\star\|_\Sigma^2 = (1+O(\rho'))\|\Sigma\|\|P_{V_{\mathrm{eff}}}w^\star\|_2^2$. On the flat V_{eff}, $w^\star|_{V_{\mathrm{eff}}} \propto a_\varepsilon|_{V_{\mathrm{eff}}}$, so by (EO3) its participation ratio is $\|P_{V_{\mathrm{eff}}}w^\star\|_1^2/\|P_{V_{\mathrm{eff}}}w^\star\|_2^2 = \Theta(r_{\mathrm{eff}})$ (Lemma 6.2, delocalization branch). Combining, $P(w^\star) = \|w^\star\|_1/\|w^\star\|_\Sigma \geq (1-O(\rho'+\beta+\rho''))\sqrt{r_{\mathrm{eff}}}/\sqrt{\|\Sigma\|}$, with equality in the aligned/flat/occupied/delocalized limit. *Failure modes.* Large $\beta \Rightarrow P(w^\star) \to \sqrt{\mathrm{tr}\,\Sigma^{-1}} \gg \sqrt{r_{\mathrm{eff}}}$; large ρ'' (subgradient pumps tail mass); violated (EO3) (spiky $a_\varepsilon|_{V_{\mathrm{eff}}}$) $\Rightarrow$ the solution sits on $O(1)$ coordinates and $P(w^\star) = O(1)$. $\square$

B.14 Proof of Corollary 8.4 (certification radius)

$R(\varepsilon; w^\star) = \Phi(t) = \frac{1}{2}$ iff $t = 0$ iff $\varepsilon\|w^\star\|_1 = \langle w^\star, \mu\rangle$, i.e. $\varepsilon_{\mathrm{cert}} = \langle w^\star, \mu\rangle / \|w^\star\|_1$. For $\varepsilon > \varepsilon_{\mathrm{cert}}$, $t > 0$ and $R > \frac{1}{2}$; under the fixed binary-channel reduction the linear discriminant then carries no information. □

C The Optimal Robust Radius

Proof of Proposition 8.5. Let $F(\varepsilon) := V(\varepsilon) - C(\varepsilon)$, so $F'(\varepsilon) = V'(\varepsilon) - C'(\varepsilon) = K|P'_{\mathrm{breach}}(\varepsilon)| - \lambda_\varepsilon$. **(i) Existence.** $F'(0^+) = K|P'_{\mathrm{breach}}(0)| - C'(0^+) = K|P'_{\mathrm{breach}}(0)| - \lambda_0$; an interior maximizer $\varepsilon^{**} > 0$ exists iff $F'(0^+) > 0$, i.e. $K|P'_{\mathrm{breach}}(0)| > \lambda_0$. The criterion is the *true* marginal cost $C'(0^+) = \lambda_0 = \lim_{\varepsilon\to 0^+} \lambda_\varepsilon$, which is an over-confident-regime quantity ($t(0) \ll 0$, $\phi(t) \to 0$) and is generally $\ll c_M\sqrt{k_\varepsilon}/\sigma$; using the margin-density floor here would be incorrect. **(ii) Uniqueness.** V' is strictly decreasing (assumption (c)) and $C' = \lambda_\varepsilon$ is nondecreasing on the relevant range (the price rises as the margin tightens), so F' is strictly decreasing and has at most one zero; combined with (i) it has exactly one, and F is strictly concave there. **(iii) Boundary condition.** At ε^{**}, $F' = 0$ gives $K|P'_{\mathrm{breach}}(\varepsilon^{**})| = \lambda_\varepsilon(\varepsilon^{**})$. **(iv) Comparative statics.** By the implicit function theorem on $F'(\varepsilon^{**}; K, k_\varepsilon) = 0$: $\partial\varepsilon^{**}/\partial K = -F'_K/F'_\varepsilon = |P'_{\mathrm{breach}}|/(-F''_\varepsilon) > 0$ (numerator > 0, $F''_\varepsilon < 0$); and, through an exogenous complexity parameter π that raises the whole marginal-cost schedule ($\partial\lambda_\varepsilon/\partial\pi > 0$, e.g. $\pi = d$), $\partial\varepsilon^{**}/\partial\pi = -F'_\pi/F'_\varepsilon < 0$ with $F'_\pi = -\partial\lambda_\varepsilon/\partial\pi < 0$; in particular $\partial\varepsilon^{**}/\partial d < 0$. We differentiate through the exogenous π, not through the endogenous $k_\varepsilon = k_\varepsilon(\varepsilon)$, which is not a free parameter that can be held fixed while ε varies. **(v)** Since $F'(\varepsilon_{\mathrm{cert}}) \le 0$ (past the certification radius the value gain vanishes while cost persists), $\varepsilon^{**} \le \varepsilon_{\mathrm{cert}}$. □

D Engineering Interpretation

D.1 An insurance analogy

The robust marginal price can be read as an insurance premium: the coverage is the adversarial radius ε, the premium is λ_ε, and the claim is the worst-case perturbation $\max_\delta \ell(x+\delta, y)$. The ℓ_∞ adversary presses on all coordinates at once, forcing the model to reserve against a claim on every effective dimension ($\sqrt{d}$, or $\sqrt{r_{\mathrm{eff}}}$)—this is the intuitive source of prefactor saturation—and coverage lapses (in the information-theoretic sense) once ε crosses the certification radius $\varepsilon_{\mathrm{cert}}$. The analogy is heuristic; the rigorous statement is the subgradient language of the main text.

D.2 The dimensional tax as a design constraint

Theorem 4.1 says the tax $\lambda_\varepsilon \ge c_M\sqrt{k_\varepsilon}/\sigma$ is dictated by the ℓ_∞/ℓ_1 geometry, not a modeling choice: for a fixed operating point one cannot reduce the marginal price below the effective-support floor by changing the discriminant, only by changing the effective support itself.

D.3 The measurement recipe

The cheap-proxy bridge (Proposition 8.1, Proposition 8.2) gives an operational recipe: estimate the robust price at small ε by $\mathbb{E}\|\nabla_x \ell\|_1$ (one backward pass), with an $O(\varepsilon^2)$ curvature correction controlled by κ_∞ (Remark 8.3).

D.4 Lowering the tax via effective rank

The lever for reducing the tax is the effective support/stable rank, as developed in Section 8, §8.5: compressing the effective rank of the representation lowers the prefactor from $\sqrt{d}$ toward $\sqrt{r_{\mathrm{eff}}}$ (conditional on effective occupancy). We refer to that section for the low-rank-adaptation and LLM readings and their order-of-magnitude estimate.

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